

Status

HelixTranslate — Feature Status

Revision: 14 **Last modified:** 2026-06-16T14:30:00Z **Authority:** Derived from docs/features/.feature_inventory_raw.md (Revision 1). Per §11.4.45 (captured-evidence Status doc), §11.4.44 (revision header), §11.4.56 (two-audience summary → Status_Summary.md), §11.4.60 (always-sync), §11.4.6 (no-guessing — every status reflects what the inventory actually found in source). **Scope:** Every one of the 417 inventoried features across CLI, API, Web, gRPC, Library, Submodule and Infra. helix_qa deliberately not included.

Anti-bluff note (§11.4.6 / §11.4.2 / §11.4.107): Video-confirmation is set to a real file **only** where a recording exists AND that recording genuinely demonstrates the feature working (each cited .mp4 was ffprobe-verified for non-zero duration+frames and content-verified against its claimed behaviour — e.g. real DeepSeek output in the target language, real gRPC translated text, live WebSocket count change). **43 feature rows are video-confirmed** (Wave 3f of 2026-06-16 added **2 net-new** — **EPUB→EPUB conversion** (helix_translate-epub-output-wave3e-20260616.mp4: EN .epub → real deepseek run → OUTPUT EPUB carries real Spanish “La Antigua Farola la vieja farola se alzaba sola en la rocosa costa...”) and **TXT→TXT translation** (helix_translate-txt-translation-wave3f-20260616.mp4: EN .txt → real deepseek run → ES .txt “El río fluía tranquilamente a través del valle verde. Los pájaros cantaban bajo la luz de la mañana.”), each ffprobe+frame-verified per §11.4.107 (790×560 yuv420p, ≥6 frames, start-frame English-only ≠ output-frame Spanish); Wave 3f also **re-confirmed** (NOT double-counted) the already-confirmed **FB2→FB2** (helix_translate-fb2-convert-wave3e-20260616.mp4, namespace preserved + real Spanish), **format-detection** (helix_translate-format-detect-wave3e-20260616.mp4, extensionless .dat detected-by-content), **-script Cyrillic↔Latin** (helix_translate-script-conversion-wave3e-20260616.mp4, RUN A latin “Ovo je testni...” ≠ RUN B cyrillic “Систем треба...”) and **markdown-translator EPUB↔MD round-trip** (helix_translate-markdown-roundtrip-wave3f-20260616.mp4, “# El Faro Viejo / El viejo faro se alzaba solitario en la costa rocosa.”) rows with fresh frame-verified clips; running total 41 (Rev 13) + 2 = **43**. Wave 3d of 2026-06-16 added **4 net-new**, prioritising feature-type diversity — the **cmd/verify-models CLI** (real LLMsVerifier discovery/scoring/selection: 14 env-keyed providers with real per-provider reachable/auth/HTTP-status/candidate/verified counts, summary total verified: 4, then model-bridge list selection ranking #1 novita/Sao10K/L3-8B-Stheno-v3.2 score=0.91875 ... #4 siliconflow score=0.88125 — helix_translate-verify-models-wave3d-20260616.mp4), the **cmd/**

workable-items CLI (real §11.4.93 SQLite SSoT: validate → OK: DB ... matches markdown (81 items), list → real ATM-NNN rows + descriptions — `helix_translate-workable-items-wave3d-20260616.mp4`), and **2 more providers** — **nvidia** (“...Un anciano pescador **caminaba** a lo largo de la orilla.” via `integrate.api.nvidia.com`, `helix_translate-provider-nvidia-translate-wave3d-20260616.mp4`) and **openrouter** (“...Un anciano pescador **paseaba** a lo largo de la orilla.” via `openrouter.ai/api`, `helix_translate-provider-openrouter-translate-wave3d-20260616.mp4`), the two provider outputs distinct (`caminaba/paseaba`) proving genuine per-provider models, all four Wave-3d clips `ffprobe+frame-verified` per §11.4.107 (790×560 yuv420p, ≥6 frames at `fps=2`, CLI clips show advancing distinct sections / provider clips show start-frame English-only ≠ output-frame Spanish); the Wave-3d providers reach the `llm_provider/OpenAI-compatible` per-provider path; upstage was tested but is an honest SKIP (`auth=false HTTP 403`), together SKIPPed (no key) — never faked. Wave 3c of 2026-06-16 added **2 net-new** — the **Hyperbolic** library client (`pkg/translator/llm`) and the **Fireworks** provider in the `llm_provider` submodule (`llm_provider/pkg/providers/fireworks/fireworks.go`), each a real-use Wave-3c .mp4 showing a distinct EN→ES translation through the OpenAI-compatible per-provider path — `helix_translate-provider-hyperbolic-translate-wave3c-20260616.mp4` (“Al amanecer, el sol salía sobre el tranquilo pueblo. Un anciano pescador caminaba por la orilla...”), and `helix_translate-provider-fireworks-translate-wave3c-20260616.mp4` (“Al amanecer, el sol ascendió sobre el tranquilo pueblo. Un anciano pescador paseaba por la orilla...”), the two outputs distinct from each other (`salía/caminaba` vs `ascendió/paseaba`) proving genuine per-provider models, `ffprobe+frame-verified` per §11.4.107 (790×560 yuv420p, ≥6 frames at `fps=2`, start-frame input-only ≠ output-frame Spanish). Wave 3c also **re-confirmed** (no double-count) the already-confirmed REST POST `/api/v1/translate` and GET `/health · /api/v1/version · /api/v1/providers` rows against a freshly-started live `cmd/server TLS` instance (port 9443) — `helix_translate-rest-translate-wave3c-20260616.mp4` (real JSON `{"translated":"El sol de la mañana se alzó sobre el tranquilo pueblo.", "provider":"llm-deepseek", ...}`) and `helix_translate-rest-endpoints-wave3c-20260616.mp4` (real `health/version/providers` JSON). The GET `/api/v1/verified-models` 404 remains an honest PENDING gap (LLMsVerifier disabled in config — not confirmed). Wave 3b of 2026-06-16 added 3 net-new — the **Gemini**, **Zhipu (GLM)** and **Cohere** library clients, each backed by a real-use Wave-3b .mp4 showing a distinct EN→SR-latin translation through the per-provider path — `helix_translate-provider-gemini-translate-wave3b-20260616.mp4` (“Staro svjetiljko stajalo je samotočno na kamenitoj obali...”), `helix_translate-provider-zhipu-translate-wave3b-20260616.mp4` (“Stara svjetiljka stojeća samostalno na kamenitoj obali...”), `helix_translate-provider-cohere-translate-wave3b-20260616.mp4` (provider=`cohere` via `-base-url https://api.cohere.ai/compatibility/v1`, “Stari svjetionik stajao je sam na kamenitoj obali... svijeća njegova vodila brodove...”), all four outputs (incl. the re-confirmed deepseek clip `helix_translate-provider-deepseek-translate-wave3b-20260616.mp4` “Stari svjetionik stajao je sam na kamenitoj obali...”) distinct from one another proving genuine per-provider models, `ffprobe-verified` (non-zero `dur+frames`, 790×560 even-dims

yuv420p) + ≥ 6 frames extracted with one content-verified per §11.4.107; the deepseek clip is a re-confirmation, no double-count; Wave 3a of 2026-06-16 added 2 net-new — the **Cerebras** and **SambaNova** per-provider LLM clients, each backed by a real-use Wave-3a .mp4 showing a distinct EN→ES translation through unified-translator -provider <name> — helix_translate-provider-cerebras-translate-wave3a-20260616.mp4 (“El sol de la mañana se alzó sobre el tranquilo pueblo.”) and helix_translate-provider-sambanova-translate-wave3a-20260616.mp4 (“El sol matutino se alzó sobre el tranquilo pueblo.”), the two outputs distinct from each other proving genuine per-provider models, ffmpeg+frame-verified per §11.4.107 with start-frame $\neq$ output-frame liveness; Wave 2 of 2026-06-16 added 5 net-new — DOCX output, PDF output, HTML→Markdown conversion, ProviderDiverseTranslators, EnsembleFactory — each backed by a real-use Wave-2 .mp4 independently validator-verified ffmpeg+frame; Wave 2 also re-confirmed the already-confirmed gRPC TranslationService row with an additional clip, no double-count; Wave 1 added 2 — bridge-ensemble best-model selection + bridge provider-diverse catalogue) across unified-translator (DeepSeek translate, EPUB→TXT / HTML→EPUB / FB2→EPUB / FB2→FB2 conversions, Serbian-Cyrillic, -script Cyrillic↔Latin, format-detection, help/version, -verify, **-multipass multi-pass polishing engine**), the REST API server (POST /api/v1/translate real DeepSeek + target_lang FIXED, TLS startup FIXED, /health · /api/v1/version · /api/v1/providers), the gRPC server (real EN→ES), markdown-translator (EPUB↔MD round-trip), preparation-translator (FIXED real analysis), cmd/cli (real DeepSeek), monitor-server (live WS hub), the DeepSeek library client, and the **LLMsVerifier→agent bridge CLI** (cmd/model-bridge — helixtranslate-bridge-bestmodel-translate-20260615.mp4: in-process verify→SQLite store→select strongest novita/Sao10K/L3-8B-Stheno-v3.2 score 0.919 → live invoke EN→ES “Good morning, friend.” → “Buenos días, amigo.”; ffmpeg 8.0s/80 frames, ≥ 6 frames content-verified) **and the LLMsVerifier→agent bridge MCP-stdio surface** (cmd/model-bridge mcp — helixtranslate-bridge-mcp-stdio-20260615.mp4: a real JSON-RPC MCP session over stdio — initialize→tools/list→bridge_best_model→bridge_invoke — where tool bridge_invoke returned a live verified-model translation “Le pont relie deux rives.” (isError:None); ffmpeg 11.88s/10 frames, ≥ 6 frames content-verified). **Now video-confirmed (Wave 2): DOCX output** (Implemented; pure-Go OOXML writer pkg/ebook/docx_writer.go, commit 87cd2be) — Wave-2 real-use video helix_translate-docx-output-20260616-112943.mp4 (unified-translator DOCX output, openai/gpt-4, the produced .docx carries real Spanish text; validator ffmpeg+frame-verified). **PDF output** (Implemented, commit fb265e7; pkg/ebook/pdf_writer.go Book→HTML5→weasyprint→Cyrillic-faithful PDF, wired into unified-translator .pdf output, main.go:1160) — Wave-2 real-use video helix_translate-pdf-output-20260616-113057.mp4 (valid PDF with real Spanish; validator ffmpeg+frame-verified). **Wave-1 bridge-translator-path blocker RESOLVED (caveat 8):** the ValidModels-whitelist runtime failure that blocked the DOCX/PDF/gRPC/per-provider videos in Wave 1 is fixed — the verified-bridge translator path now produces real per-provider translations (Wave-2 provider clips: novita/mistral/groq/siliconflow). **Honest gaps recorded, NOT confirmed:** cmd/translator + translate-ssh + ebook-translator are **Obsolete** (→ **Fixed.md**), **Rev 7** (feature-removed — the local-runtime + SSH-local

paths were removed in bridge phase-2 R-2..R-4; the default path now sources the LLMsVerifier pkg/bridge; the earlier STUB/OPERATOR-BLOCKED wording is stale and corrected); GET /api/v1/verified-models returns 404 because LLMsVerifier is disabled in config. Everything else is PENDING (a recording is owed) or N/A (no user-visible video applies — e.g. CLI flags, internal library types). Absence of a runtime recording is stated honestly, never papered over.

Operator-blocked items (read first, §11.4.45)

Two binaries are OPERATOR-BLOCKED — both require a remote SSH worker host running llama.cpp, which is not available in this environment. The binaries build and run as far as the missing host allows (captured in helixtranslate-cmd-blocked-binaries_20260615-172456.mp4):

- **cmd/translate-ssh** — SSH worker 4-step FB2→EPUB translator. **Unblock condition:** a reachable SSH host with llama.cpp + a model.
- **cmd/ebook-translator** — FB2 remote-translate workflow (syncs binary to remote, runs remote translate, downloads outputs). **Unblock condition:** same remote SSH/llama.cpp host.
- **cmd/translator (remote path)** — remote SSH/llama.cpp translate path is OPERATOR-BLOCKED (same host requirement); its LOCAL path is a separate STUB (“local translation not yet implemented”).

Coverage summary

Metric	Value
Headline feature total (inventory’s deduplicated per-category tally)	417
Enumerated feature rows in this doc	496 (Rev 14 = 494 Rev 13 + 2 new unified-translator rows added Wave 3f for the net-new videos: EPUB→EPUB conversion + TXT→TXT translation; Rev 9 = 493 Rev 8 + 1 new unified-translator HTML→Markdown conversion row added that wave for the Wave-2 html-to-md video; §11.4.6 — re-derive with awk over ^\ 9-column rows. The earlier “478” was NOT re-derivable from the live table — it was a hand-tracked running tally that had drifted; Rev 8 added the 7 missing pkg/bridge ensemble-seam rows: BestTranslator, BestTranslatorFunc, EnsembleFactory, ProviderDiverseTranslators, ProviderDiverseClients, ProviderDiverseModels, BestClient)
Implemented	445 (Rev 14: 443 Rev 13 + 2 new unified-translator rows EPUB→EPUB conversion + TXT→TXT translation, both

Metric	Value
	Implemented + Wired + Wave-3f video-confirmed; Rev 9: 442 Rev 8 + 1 new unified-translator HTML→Markdown conversion row; Rev 8 was 435 Rev 7 + 7 new pkg/bridge ensemble-seam rows; was 471 Rev 6, -36 flipped to Obsolete Rev 7 — local-runtime + SSH-local features removed in bridge phase-2 R-2..R-4)
Obsolete (→ Fixed.md)	39 (Rev 7 — bridge phase-2 R-2..R-4 removals: cmd/translate-ssh ×9, cmd/ssh-translation ×2, cmd/translator ×19, cmd/ebook-translator ×3, pkg/sshworker ×2, pkg/modelsbridge ×1, Ollama provider ×1, llama.cpp provider+coordinator ×2; Reason=feature-removed; the KEPT distributed-API path is NOT Obsolete)
Stub	4 (POST /api/v1/translate/ebook, POST /api/v1/preparation/analyze, GET /api/v1/preparation/result/:id, POST /api/v1/translate-with-verification)
Not implemented (gap rows)	0 (PDF-write flipped gap→Implemented Rev 4, commit fb265e7; output now EPUB/FB2/TXT/HTML/MD/DOCX/PDF)
Partial	2 (POST /api/batch on standalone pkg/api/server.go — returns queued batch_id only, no translation; vision_engine OpenCV — stub default, real impl behind build tag) — was 3 Rev 6; cmd/translator flipped to Obsolete Rev 7
Operator-blocked	0 (was 2 Rev 6 — translate-ssh + ebook-translator flipped to Obsolete Rev 7: the remote SSH/llama.cpp worker capability was removed, not merely host-blocked)
Video-confirmed	43 (real, content-verified recordings — see Anti-bluff note; Rev-14 / Wave 3f of 2026-06-16 added 2 net-new video confirmations prioritising feature-type diversity — EPUB→EPUB conversion (helix_translate-epub-output-wave3e-20260616.mp4, EN .epub → real Spanish EPUB) and TXT→TXT translation (helix_translate-txt-translation-wave3f-20260616.mp4, EN .txt → real Spanish .txt), each ffprobe+frame-verified per §11.4.107; Wave 3f also re-confirmed (NOT double-counted) FB2→FB2 / format-detection /

Metric	Value
	–script Cyrillic[?]Latin / markdown Epub[?]MD round-trip with fresh frame- verified clips; running total 41 (Rev 13) + 2 = 43. Rev-13 / Wave 3d of 2026-06-16 added 4 net-new video confirmations prioritising feature-type diversity — the cmd/verify-models CLI (helix_translate-verify-models- wave3d-20260616.mp4, real LLMsVerifier discovery/scoring/selection + ranking), the cmd/workable-items CLI (helix_translate-workable- items-wave3d-20260616.mp4, real §11.4.93 SQLite validate+list), and 2 more providers nvidia (helix_translate- provider-nvidia-translate- wave3d-20260616.mp4, “...caminaba a lo largo de la orilla.”) + openrouter (helix_translate-provider- openrouter-translate- wave3d-20260616.mp4, “...paseaba a lo largo de la orilla.”), each ffprobe+frame- verified per §11.4.107; running total 37 (Rev 12) + 4 = 41. upstage HTTP-403 SKIP + together no-key SKIP, never faked. Rev-12 / Wave 3c of 2026-06-16 added 2 net-new video confirmations — the Hyperbolic library client (helix_translate-provider- hyperbolic-translate- wave3c-20260616.mp4, “...el sol salía... caminaba por la orilla...” and the Fireworks llm_provider submodule client (helix_translate-provider- fireworks-translate- wave3c-20260616.mp4, “...el sol ascendió...paseaba por la orilla...”), each a distinct real EN→ES translation via the OpenAI-compatible per-provider path, ffprobe+frame-verified per §11.4.107 (790×560 yuv420p, ≥6 frames at fps=2, start-frame input-only ≠ output-frame Spanish), the two outputs distinct proving genuine per-provider models; Wave 3c also re-confirmed (NOT double-counted) the already-confirmed REST POST /api/v1/ translate+ /health · /api/v1/ version · /api/v1/providers rows against a live cmd/server TLS instance (helix_translate-rest-translate- wave3c-20260616.mp4 + helix_translate-rest-endpoints- wave3c-20260616.mp4); count

Metric	Value
	reconciliation (§11.4.6): this cell read 32 through Rev 10/11 — a sync lag behind the Anti-bluff-note prose; it is corrected to the true running total here — 30 (Rev 9) + 2 (Wave 3a Cerebras/SambaNova) + 3 (Wave 3b Gemini/Zhipu/Cohere) + 2 (Wave 3c Hyperbolic/Fireworks) = 37. Rev-10 / Wave 3a of 2026-06-16 added 2 net-new video confirmations — the Cerebras and SambaNova per-provider LLM clients (helix_translate-provider-cerebras-translate-wave3a-20260616.mp4 + helix_translate-provider-sambanova-translate-wave3a-20260616.mp4), each a distinct real EN→ES translation via unified-translator -provider <name>, ffmpeg+frame-verified per §11.4.107 with start-frame ≠ output-frame liveness, the two outputs distinct proving genuine per-provider models; Rev-11 / Wave 3b of 2026-06-16 added 3 net-new — the Gemini, Zhipu (GLM) and Cohere library clients (helix_translate-provider-{gemini,zhipu,cohere}-translate-wave3b-20260616.mp4), distinct EN→SR-latin per-provider translations, frame-verified per §11.4.107; Rev-9 / Wave 2 of 2026-06-16 added 5 net-new video confirmations, each backed by a real-use Wave-2 .mp4 independently validator-verified ffmpeg+frame §11.4.107: DOCX output (helix_translate-docx-output-20260616-112943.mp4), PDF output (helix_translate-pdf-output-20260616-113057.mp4), HTML→Markdown conversion (helix_translate-html-to-md-20260616-114228.mp4), ProviderDiverseTranslators + EnsembleFactory (the 4 per-provider clips novita/mistral/groq/siliconflow — helix_translate-provider-{novita,mistral,groq,siliconflow}-translate-20260616-114*.mp4). Wave 2 also re-confirmed the already-confirmed gRPC TranslationService row with an additional clip helix_translate-grpc-translate-20260616-113415.mp4 (NOT double-counted — that row was already video-confirmed). DOCX output

Metric	Value
	and PDF output are no longer “real-artifact-only” — they are now video-confirmed. §11.4.108 note: the Wave-1 bridge-translator-path runtime blocker (caveat 8) is RESOLVED, which is what made the DOCX/PDF/gRPC/per-provider Wave-2 recordings genuinely demonstrate working features (recording them in Wave 1 would have been a §11.4.107 bluff; they were honest SKIPS then). Wave 1 added 2: bridge-ensemble best-model selection + bridge provider-diverse catalogue. $\approx 8.3\%$ of 494 enumerated rows ($\approx 9.8\%$ of the 417 headline)
Video-confirmation PENDING	runtime/user-visible features without a recording yet
Video-confirmation N/A	flags, internal library types, infra middleware (no standalone user-visible video)

§11.4.6 reconciliation: The inventory’s headline is **417** — its own note states this is an *approximate, deduplicated* per-category tally where “some rows are sub-features... counted once, and a few rows span two categories (counted once, in the first/primary category).” The inventory’s *detailed* tables actually enumerate **470** feature rows. This document transcribes **all 470 detailed rows 1:1** so **no feature is dropped** (which would itself be a bluff). The 417 headline and the 470 enumerated rows are both reported honestly; the 53-row gap is dedup, not missing coverage.

“Static-return” endpoints (e.g. GET /api/stats hardcoded zeros, GET /api/v1/status/:id hardcoded completed, GET /api/v1/metrics static, monitor-server GET /api/v1/status static monitoring_active) are marked **Implemented** at the route level but their Real-use cell states the static/hardcoded behaviour honestly so no false “working backend” impression is created.

Per-category counts

Category	Count
CLI	130
API	70
Web	23
gRPC	7
Library	81
Submodule	95
Infra	11
TOTAL	417

Legend

- **Implementation:** Implemented (real code path present) / Stub (returns mock, no real work) / Partial (does some but not the named work).
- **Wiring:** Wired (reachable from a binary / route / public API) / Unwired (present but not reached).
- **Real-use:** what the code actually does at runtime (honest, §11.4.6).
- **Tests:** Covered (unit/integration/challenge present in repo) / None (no test found / not asserted in inventory).
- **Validation:** strongest evidence available.
- **Video-confirmation:** Confirmed:<file> only with a real recording; else PENDING / N/A.

Per §11.4.6, where the inventory did not read test files line-by-line, Tests is recorded Not-inventoried (honest unknown) rather than guessed as Covered or None. The repo is known to carry extensive `_test.go` suites and a `test/` tree, but per-feature test attribution was not part of the raw inventory.

1. CLI Binaries — cmd/

cmd/api-server

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	V
api-server	REST translation server	API/Web	Implemented	Wired	Gin HTTP server (:8080) proxying to gRPC backend; serves dashboard + static files	Not-inventoried	S co (c se m
api-server	-http-port flag	API	Implemented	Wired	Override HTTP port (else HTTP_PORT env)	Not-inventoried	S co
api-server	-http-address flag	API	Implemented	Wired	Override HTTP bind address (else HTTP_ADDRESS env)	Not-inventoried	S co
api-server	-grpc-address flag	API	Implemented	Wired	Override gRPC backend address (else	Not-inventoried	S co

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Verdict
api-server	-debug flag	API	Implemented	Wired	GRPC_ADDRESS env) Enable debug / Gin debug mode	Not-inventoried	Score 100

cmd/challenge-runner

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Verdict
challenge-runner	Challenge orchestrator	CLI	Implemented	Wired	Registers + runs all anti-bluff challenges, prints PASS/ FAIL/ SKIP, writes results dir	Not-inventoried	Score 100
challenge-runner	-parallel	CLI	Implemented	Wired	Run challenges in parallel	Not-inventoried	Score 100
challenge-runner	-max-concurrency	CLI	Implemented	Wired	Max concurrent challenges	Not-inventoried	Score 100
challenge-runner	-stop-on-failure	CLI	Implemented	Wired	Stop on first failure	Not-inventoried	Score 100
challenge-runner	-verbose	CLI	Implemented	Wired	Verbose output	Not-inventoried	Score 100
challenge-runner	-timeout	CLI	Implemented	Wired	Per-challenge timeout	Not-inventoried	Score 100
challenge-runner	-filter	CLI	Implemented	Wired	Comma-separated challenge IDs to run	Not-inventoried	Score 100
challenge-runner	-category	CLI	Implemented	Wired	Limit to a category	Not-inventoried	Score 100
challenge-runner	-results-dir	CLI	Implemented	Wired	Results output directory	Not-inventoried	Score 100

cmd/cli

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Value
cli	Universal ebook translator	CLI	Implemented	Wired	Parse ebook → translate → atomic-write EPUB/FB2/TXT; multi-llm/distributed support	Not-inventoried	Parse ebook → translate → atomic-write EPUB/FB2/TXT; multi-llm/distributed support (default: sa, W (8) va w sil os su pa G no (5) pr
cli	-i/-input	CLI	Implemented	Wired	Input ebook (FB2/EPUB/TXT/HTML/PDF/DOCX)	Not-inventoried	Sc
cli	-o/-output	CLI	Implemented	Wired	Output file	Not-inventoried	Sc
cli	-f/-format	CLI	Implemented	Wired	Output format (epub, fb2, txt)	Not-inventoried	Sc
cli	-p/-provider	CLI	Implemented	Wired	Translation provider	Not-inventoried	Sc
cli	-model	CLI	Implemented	Wired	LLM model name	Not-inventoried	Sc
cli	-api-key	CLI	Implemented	Wired	Provider API key	Not-inventoried	Sc
cli	-base-url	CLI	Implemented	Wired	Provider base URL	Not-inventoried	Sc
cli	-script	CLI	Implemented	Wired	Output script (default/cyrillic/latin)	Not-inventoried	Sc
cli	-locale	CLI	Implemented	Wired			Sc

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Ver
					Target language locale code	Not-inventoried	
cli	-language	CLI	Implemented	Wired	Target language name	Not-inventoried	Sc
cli	-source	CLI	Implemented	Wired	Source language (auto-detect if unset)	Not-inventoried	Sc
cli	-detect	CLI	Implemented	Wired	Detect source language and exit (LLM detector if API key)	Not-inventoried	Sc
cli	-v/-version	CLI	Implemented	Wired	Show version	Not-inventoried	Sc
cli	-h/-help	CLI	Implemented	Wired	Show help	Not-inventoried	Sc
cli	-create-config	CLI	Implemented	Wired	Write a config-file template	Not-inventoried	Sc
cli	-disable-local-llms	CLI	Implemented	Wired	Use only distributed/API providers	Not-inventoried	Sc
cli	-prefer-distributed	CLI	Implemented	Wired	Prefer distributed workers	Not-inventoried	Sc
cli	-c/-config	CLI	Implemented	Wired	Config file path	Not-inventoried	Sc
cli	-hash-codebase	CLI	Implemented	Wired	Compute codebase hash and exit	Not-inventoried	Sc

cmd/deployment

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
deployment	Distributed deployment manager	CLI	Implemented	Wired	Orchestrates Docker-based main+worker deployment over SSH	Not-invent

Not-inventoried | Source-confirmed | N/A | | ebook-translator | Output verification | CLI | **Obsolete (→ Fixed.md)** | Wired | Verifies MD/EPUB exist, non-degenerate, target-language (Cyrillic) check | Not-inventoried | Source-confirmed | PENDING |

cmd/grpc-server

Component	Feature	Category	Implementation	Wiring	Real-use	
grpc-server	gRPC TranslationService	gRPC	Implemented	Wired	gRPC server (:50051): Start/Status/List/Cancel/Stream/GetProviders/SubscribeEvents	M i
grpc-server	-address	gRPC	Implemented	Wired	Bind address (or GRPC_ADDRESS env)	M i
grpc-server	-port	gRPC	Implemented	Wired	Port (or GRPC_PORT env)	M i
grpc-server	-max-connections	gRPC	Implemented	Wired	Max concurrent translations	M i
grpc-server	-reflection	gRPC	Implemented	Wired	Enable gRPC reflection (or ENABLE_REFLECTION env)	M i
grpc-server	-metrics	gRPC	Implemented	Wired	Enable metrics (or ENABLE_METRICS env)	M i
grpc-server	-log-level	gRPC	Implemented	Wired	Log level (or LOG_LEVEL env)	M i
grpc-server	-version	gRPC	Implemented	Wired	Show version	M i
grpc-server	-help	gRPC	Implemented	Wired	Show help	M i

cmd/markdown-translator

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
markdown-translator	EPUB [?] Markdown translate pipeline	CLI	Implemented	Wired	EPUB/MD → markdown → translate → EPUB/ MD; optional preparation phase	Not-inventoried
markdown-translator	-input	CLI	Implemented	Wired	Input file (EPUB or Markdown)	Not-inventoried
markdown-translator	-output	CLI	Implemented	Wired	Output file (auto if empty)	Not-inventoried
markdown-translator	-format	CLI	Implemented	Wired	Output format (epub, md)	Not-inventoried
markdown-translator	-lang	CLI	Implemented	Wired	Target language code	Not-inventoried
markdown-translator	-provider	CLI	Implemented	Wired	LLM provider (deepseek/ openai/ anthropic/ llamacpp)	Not-inventoried
markdown-translator	-model	CLI	Implemented	Wired	LLM model	Not-inventoried
markdown-translator	-keep-md	CLI	Implemented	Wired	Keep intermediate markdown	Not-inventoried
	-prepare	CLI	Implemented	Wired		

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
markdown-translator					Enable multi-LLM preparation/analysis phase	Not-inventor
markdown-translator	-prep-passes	CLI	Implemented	Wired	Number of preparation passes	Not-inventor

cmd/monitor-server

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
monitor-server	WebSocket monitoring hub	Web	Implemented	Wired	Gin server (:8090, MONITOR_SERVER_/ws, /monitor dashboard status + health	
monitor-server	GET /ws	Web	Implemented	Wired	WebSocket subscription by session_id	
monitor-server	/monitor + /	Web	Implemented	Wired	Serve monitor.html (redirect from /)	
monitor-server	GET /api/v1/status/:session_id	Web/API	Implemented	Wired	Returns static monitoring_action JSON (not a live session lookup)	
monitor-server	GET /health	Web	Implemented	Wired	Health + client count	
monitor-server	MONITOR_SERVER_PORT env	Web	Implemented	Wired	Configure bind port (CLI flag)	

cmd/preparation-translator

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
preparation-translator	Preparation+translation runner	CLI	Implemented	Wired	Parse ebook → multi-LLM preparation analysis → translate → write EPUB + analysis JSON	Not-inventor

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
preparation-translator	-input	CLI	Implemented	Wired	Input ebook path	Not-invented
preparation-translator	-output	CLI	Implemented	Wired	Output EPUB path	Not-invented
preparation-translator	-analysis	CLI	Implemented	Wired	Preparation analysis JSON output path	Not-invented
preparation-translator	-source	CLI	Implemented	Wired	Source language	Not-invented
preparation-translator	-target	CLI	Implemented	Wired	Target language	Not-invented
preparation-translator	-passes	CLI	Implemented	Wired	Number of preparation passes	Not-invented
preparation-translator	-providers	CLI	Implemented	Wired	Comma-separated LLM providers	Not-invented

cmd/server

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
server	Main TLS HTTP/2+HTTP/3 server	API/Web	Implemented	Wired	TLS REST/WebSocket API (config.json); CORS + rate limit + JWT; optional distributed + LLMsVerifier	Not-inventoried
server	-config	API	Implemented	Wired	Config file path (default config.json)	Not-inventoried
server	-version	API	Implemented	Wired	Show version	Not-inventoried
server	-generate-certs	CLI	Implemented	Wired	Generate self-signed TLS certs	Not-inventoried

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
					(0600 key) and exit	
server	HTTP/3 (QUIC) + HTTP/2 fallback serving	API/Web	Implemented	Wired	Serve API over TLS; HTTP/3 if EnableHTTP3	Not-inventoried
server	API routes (pkg/api Handler)	API/Web	Implemented	Wired	Registers translation REST + WebSocket routes	Not-inventoried
server	LLMsVerifier routes (if enabled)	API	Implemented	Wired	Registers verifier endpoints	Not-inventoried

cmd/ssh-translation

Obsolete-Details: Since=bridge phase-2 R-2/R-4 (2026-06-16); Reason=feature-removed (local-runtime + SSH-local translation removed; default path now sources the LLMsVerifier bridge — no local llama.cpp/Ollama, no SSH-local worker); Superseding-item=pkg/bridge (LLMsVerifier→agent bridge); Operator-approval=2026-06-15 D1/D2 (forbid all local runtimes) + R-1 redirect + the R-4 “keep distributed API, remove SSH-local” decision (2026-06-16); Triple-check evidence=source files removed (FACT: git ls-files empty for the removed cmd/* dirs + pkg/sshworker + pkg/modelsbridge + ollama.go/llamacpp*.go) + CM-NO-LOCAL-RUNTIME gate PASS. The KEPT distributed-API path is NOT Obsolete. | Component | Feature | Category | Implementation | Wiring | Real-use | Tests | Validation | Video-confirmation | |ssh-translation | SSH translation system (fixed-config) | CLI/Web | **Obsolete** (→ **Fixed.md**) | Wired | Hardcoded Russian→Serbian-Cyrillic workflow over SSH worker w/ llama.cpp; WebSocket monitoring | Not-inventoried | Source-confirmed | PENDING | | ssh-translation | (no CLI flags) | CLI | **Obsolete** (→ **Fixed.md**) | Wired | All config hardcoded in SystemConfig (host, creds, langs, llama providers) | Not-inventoried | Source-confirmed | N/A |

cmd/translate-ssh

Obsolete-Details: Since=bridge phase-2 R-2/R-4 (2026-06-16); Reason=feature-removed (local-runtime + SSH-local translation removed; default path now sources the LLMsVerifier bridge — no local llama.cpp/Ollama, no SSH-local worker); Superseding-item=pkg/bridge (LLMsVerifier→agent bridge); Operator-approval=2026-06-15 D1/D2 (forbid all local runtimes) + R-1 redirect + the R-4 “keep distributed API, remove SSH-local” decision (2026-06-16); Triple-check evidence=source files removed (FACT: git ls-files empty for the removed cmd/* dirs +

Not-inventoried | Source-confirmed | N/A | | translator | `-ssh-user` | CLI | **Obsolete** ($\rightarrow$ **Fixed.md**) | Wired | SSH username | Not-inventoried | Source-confirmed | N/A | | translator | `-ssh-password` | CLI | **Obsolete** ($\rightarrow$ **Fixed.md**) | Wired | SSH password | Not-inventoried | Source-confirmed | N/A | | translator | `-ssh-port` | CLI | **Obsolete** ($\rightarrow$ **Fixed.md**) | Wired | SSH port | Not-inventoried | Source-confirmed | N/A | | translator | `-remote-dir` | CLI | **Obsolete** ($\rightarrow$ **Fixed.md**) | Wired | Remote working directory | Not-inventoried | Source-confirmed | N/A | | translator | `-workers` | CLI | **Obsolete** ($\rightarrow$ **Fixed.md**) | Wired | Parallel workers | Not-inventoried | Source-confirmed | N/A | | translator | `-chunk-size` | CLI | **Obsolete** ($\rightarrow$ **Fixed.md**) | Wired | Text chunk size | Not-inventoried | Source-confirmed | N/A | | translator | `-concurrency` | CLI | **Obsolete** ($\rightarrow$ **Fixed.md**) | Wired | Max concurrent operations | Not-inventoried | Source-confirmed | N/A | | translator | `-verify` | CLI | **Obsolete** ($\rightarrow$ **Fixed.md**) | Wired | Verify translated output | Not-inventoried | Source-confirmed | N/A | | translator | `-verbose` | CLI | **Obsolete** ($\rightarrow$ **Fixed.md**) | Wired | Verbose logging | Not-inventoried | Source-confirmed | N/A | | translator | `-llama-binary` | CLI | **Obsolete** ($\rightarrow$ **Fixed.md**) | Wired | Path to llama.cpp binary | Not-inventoried | Source-confirmed | N/A | | translator | `-temperature` | CLI | **Obsolete** ($\rightarrow$ **Fixed.md**) | Wired | LLM temperature | Not-inventoried | Source-confirmed | N/A | | translator | `-context` | CLI | **Obsolete** ($\rightarrow$ **Fixed.md**) | Wired | LLM context size | Not-inventoried | Source-confirmed | N/A | | translator | `-version` | CLI | **Obsolete** ($\rightarrow$ **Fixed.md**) | Wired | Show version | Not-inventoried | Source-confirmed | N/A | | translator | `-help` | CLI | **Obsolete** ($\rightarrow$ **Fixed.md**) | Wired | Show help | Not-inventoried | Source-confirmed | N/A | | translator | `-hash-codebase` | CLI | **Obsolete** ($\rightarrow$ **Fixed.md**) | Wired | Compute codebase hash | Not-inventoried | Source-confirmed | N/A |

cmd/unified-translator (PRIMARY CLI)

Component	Feature	Category	Implementation	Wiring	Real-use	Test
unified-translator	Primary multi-provider translator	CLI	Implemented	Wired	Parse $\rightarrow$ MD $\rightarrow$ translate (API/llamacpp/ssh) $\rightarrow$ script-normalize $\rightarrow$ output (epub/fb2/html/txt/md) + session report	Not-inve
unified-translator	DeepSeek translate (provider=deepseek)	CLI	Implemented	Wired	EN $\rightarrow$ Serbian DeepSeek run, real LLM output written to TXT	Not-inve
unified-translator	EPUB $\rightarrow$ TXT conversion	CLI	Implemented	Wired	EPUB parse $\rightarrow$ DeepSeek translate $\rightarrow$ TXT writer, real Serbian text	Not-inve

Component	Feature	Category	Implementation	Wiring	Real-use	Test
unified-translator	HTML→EPUB conversion	CLI	Implemented	Wired	HTML parse → DeepSeek translate → EPUB writer	Not-inve
unified-translator	HTML→Markdown conversion	CLI	Implemented	Wired	HTML parse → translate (openai/gpt-4 via verified bridge) → Markdown (.md) writer; real Spanish md output	Not-inve
unified-translator	FB2→EPUB conversion	CLI	Implemented	Wired	FB2 parse → DeepSeek translate → EPUB writer	Not-inve
unified-translator	FB2→FB2 conversion	CLI	Implemented	Wired	FB2 parse → DeepSeek translate → FB2 writer (namespaces preserved)	Not-inve
unified-translator	Serbian Cyrillic output	CLI	Implemented	Wired	Translate with Serbian-Cyrillic target output	Not-inve
unified-translator	EPUB→EPUB conversion	CLI	Implemented	Wired	EPUB parse → DeepSeek translate → EPUB writer (round-trip, real translated text in produced .epub)	Not-inve
unified-translator	TXT→TXT translation	CLI	Implemented	Wired	Plain-text parse → DeepSeek translate → TXT writer	Not-inve

Component	Feature	Category	Implementation	Wiring	Real-use	Test
					(real translated text, no format conversion)	
unified-translator	Format detection	CLI	Implemented	Wired	Input-format auto-detection on real input	Not-inve
unified-translator	help/version output	CLI	Implemented	Wired	-help/-version usage banner	Not-inve
unified-translator	-i/-input	CLI	Implemented	Wired	Input ebook	Not-inve
unified-translator	-o/-output	CLI	Implemented	Wired	Output file (auto if empty)	Not-inve
unified-translator	-source-lang	CLI	Implemented	Wired	Source language	Not-inve
unified-translator	-target-lang	CLI	Implemented	Wired	Target language	Not-inve
unified-translator	-script (Serbian Cyrillic↔Latin)	CLI	Implemented	Wired	Target script (cyrillic/latin) script normalization on real run	Not-inve
unified-translator	-provider	CLI	Implemented	Wired	Provider (openai/anthropic/zhipu/deepseek/	Not-inve

Component	Feature	Category	Implementation	Wiring	Real-use	Test
					qwen/gemini/ ollama/ llamacpp/ssh)	
unified-translator	-model	CLI	Implemented	Wired	Model name	Not-inve
unified-translator	-api-key	CLI	Implemented	Wired	Provider API key	Not-inve
unified-translator	-base-url	CLI	Implemented	Wired	Provider base URL	Not-inve
unified-translator	-temperature	CLI	Implemented	Wired	LLM temperature	Not-inve
unified-translator	-max-tokens	CLI	Implemented	Wired	Max tokens	Not-inve
unified-translator	-timeout	CLI	Implemented	Wired	Request timeout	Not-inve
unified-translator	-ssh-host	CLI	Implemented	Wired	SSH host (provider=ssh)	Not-inve
unified-translator	-ssh-user	CLI	Implemented	Wired	SSH username	Not-inve
unified-translator	-ssh-password	CLI	Implemented	Wired	SSH password	Not-inve
unified-translator	-ssh-port	CLI	Implemented	Wired	SSH port	Not-inve
unified-translator	-remote-dir	CLI	Implemented	Wired	Remote directory	Not-inve
unified-translator	-llama-binary	CLI	Implemented	Wired	Path to llama.cpp binary	Not-inve
unified-translator	-llama-model	CLI	Implemented	Wired	Path to llama.cpp model	Not-inve
unified-translator	-context-size	CLI	Implemented	Wired	LLM context size	Not-inve
unified-translator	-workers	CLI	Implemented	Wired	Parallel workers	Not-inve
unified-translator	-chunk-size	CLI	Implemented	Wired	Text chunk size	Not-inve
unified-translator	-concurrency	CLI	Implemented	Wired	Concurrent operations	Not-inve
unified-translator	-verify (output-quality check)	CLI	Implemented	Wired	Built-in per-step verification: input/markdown/translation-quality/valid-EPUB verdicts	Not-inve

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
					in session report. NOTE: this is the unified-translator's own verify pass, NOT the pkg/verification multipass engine (that engine is not wired into this CLI — see gap row below)	
unified-translator	-verbose	CLI	Implemented	Wired	Verbose logging	Not-invented
unified-translator	-monitoring	CLI/Web	Implemented	Wired	Enable web monitoring	Not-invented
unified-translator	-monitoring-port	CLI/Web	Implemented	Wired	Monitoring server port	Not-invented
unified-translator	-use-verifier	CLI	Implemented	Wired	Use LLMsVerifier for model selection	Not-invented
unified-translator	-verifier-url	CLI	Implemented	Wired	LLMsVerifier API URL	Not-invented
unified-translator	-verifier-api-key	CLI	Implemented	Wired	LLMsVerifier API key	Not-invented
unified-translator	-version	CLI	Implemented	Wired	Show version	Not-invented
unified-translator	-help	CLI	Implemented	Wired	Show help	Not-invented

cmd/verify-models

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Verdict
verify-models	In-process LLMsVerifier runner	CLI	Implemented	Wired	Runs verification pipeline against env-keyed providers; persists verified models to SQLite	Not-inventoried	Passes LLM security case verification - no pipeline key provided - restate

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Ver
							co
							re
							au
							ca
							op
							st
							au
							st
							si
							ve
							su
							ve
							pe
							th
							li
							se
							no
							St
							so
							si
							09
							so
							W
							as
							ffj
							yu
							fp
							ve
verify-models	-db	CLI	Implemented	Wired	SQLite store path	Not-inventoried	So
verify-models	-max	CLI	Implemented	Wired	Max models/provider through pipeline	Not-inventoried	So
verify-models	-min-score	CLI	Implemented	Wired	Min overall score to persist	Not-inventoried	So
verify-models	-timeout	CLI	Implemented	Wired	Overall verification timeout	Not-inventoried	So

cmd/model-bridge

LLMsVerifier→component+agent bridge (no local models): selects the best LLMsVerifier-scored model, exposes it to agents/components via a CLI and an MCP stdio server wired in `.mcp.json`. Source: `cmd/model-bridge/{main.go,mcp.go}`, `pkg/bridge/`, commit `ab1bed3` (+ routing seam `a5860b2`).

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
model-bridge	LLMsVerifier→agent bridge CLI	CLI	Implemented	Wired	Selects best LLMsVerifier-scored model (top-1 + fallback chain) and translates/invokes through it; no local models	Coverage of bridge unit-tests go to green session
model-bridge	MCP stdio server (.mcp.json)	CLI/Infra	Implemented	Wired	Exposes the selected model to MCP clients over stdio; wired as model-bridge in .mcp.json (./build/model-bridge)	Coverage of model-mcp_

cmd/workable-items

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
workable-items	§11.4.93 workable-items	CLI	Implemented	Wired	Bidirectional sync between Issues/Fixed markdown and	Not-inventoried

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
	SQLite tool				SQLite single-source-of-truth	
workable-items	sync md-to-db	CLI	Implemented	Wired	Parse ATM-NNN headings from markdown into DB	Not-inventoried
workable-items	sync db-to-md	CLI	Implemented	Wired	Regenerate Issues_Summary/Fixed_Summary from DB	Not-inventoried
workable-items	validate	CLI	Implemented	Wired	Verify DB matches markdown; exit non-zero on drift	Not-inventoried
workable-items	diff	CLI	Implemented	Wired	Print md-vs-db drift human-readably	Not-inventoried
workable-items	list	CLI	Implemented	Wired	Print items in DB	Not-inventoried
workable-items	record-event	CLI	Implemented	Wired	Append event to item_history audit trail	Not-inventoried
workable-items	path/ value flags	CLI	Implemented	Wired	-issues/-fixed/-db/-issues-summary/-fixed-summary/-atm/-event/-on/-not	Not-inventoried

2. Library Packages — pkg/

pkg/translator/llm — LLM provider clients & factory

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
LLM Core	LLMClient interface & factory	Library	Implemented	Wired	LLMClient interface + NewLLMTranslator factory dispatching to ~32 providers	Not-inventoried
LLM Core	Type-coercion utilities	Library	Implemented	Wired	JSON config conversion helpers	Not-inventoried
LLM Core	Verified factory	Library	Implemented	Wired	LLMsVerifier-gated translator construction w/ fallback chain (CONST-034)	Not-inventoried
LLM Core	Mock client	Library	Implemented	Wired	Test mock LLMClient with fault injection (test-only)	Not-inventoried
OpenAI	OpenAI client	Library	Implemented	Wired	GPT chat-completions translation	Not-inventoried
Anthropic	Anthropic Claude client	Library	Implemented	Wired	Claude /v1/ messages translation	Not-inventoried
Google	Gemini client	Library	Implemented	Wired	Google Gemini generativelanguage API	Not-inventoried

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
DeepSeek	DeepSeek client	Library	Implemented	Wired	DeepSeek OpenAI-compatible chat-completions	Not-inventori
Zhipu	Zhipu (GLM) client	Library	Implemented	Wired	Zhipu GLM /api/paas/v4	Not-inventori
Alibaba	Qwen client	Library	Implemented	Wired	Qwen/DashScope (OAuth2-capable)	Not-inventori
Ollama	Ollama local client	Library	Obsolete (→ Fixed.md)	Wired	Local /api/generate inference — REMOVED (bridge phase-2 R-2). See Obsolete-Details below.	Not-inventori
llama.cpp	llama.cpp client	Library	Obsolete (→ Fixed.md)	Wired	llama-cli wrapper w/ hardware detection, model download, GPU	Not-inventori

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
llama.cpp	llama.cpp multi-worker coordinator	Library	Obsolete (→ Fixed.md)	Wired	accel — REMOVED (bridge phase-2 R-3). See Obsolete-Details below. Worker pool for parallel local llama.cpp inference — REMOVED (bridge phase-2 R-3). See Obsolete-Details below.	Not-inventori

Obsolete-Details (Ollama + llama.cpp provider/coordinator rows above):
 Since=bridge phase-2 R-2/R-3 (2026-06-16); Reason=feature-removed (local-runtime inference removed; the default path now sources the LLMsVerifier bridge — no local llama.cpp/Ollama); Superseding-item=pkg/bridge; Operator-approval=2026-06-15 D1/D2 (forbid all local runtimes) + R-1 redirect; Triple-check evidence=pkg/translator/llm/{ollama.go, llamacpp.go, llamacpp_provider.go} removed (FACT: git ls-files empty) + CM-N0-LOCAL-RUNTIME gate PASS. The KEPT API provider clients (OpenAI/Anthropic/DeepSeek/...) in this section are NOT Obsolete. | Groq | Groq client | Library | Implemented | Wired | OpenAI-compatible wrapper | Not-inventoried | Source-confirmed | PENDING || Cohere | Cohere client | Library | Implemented | Wired | command-r via OpenAI-compat endpoint | Not-inventoried | **PASS — real Cohere translation captured** (Wave-3b real-use video: unified-translator -provider cohere -model command-r-08-2024 -base-url https://api.cohere.ai/compatibility/v1, real key, EN→SR-latin “Stari svjetionik stoejo je sam na kamenitoj obali. Noću svaki put svijeća njegova vodila brodove sigurno kući kroz oluju.”; log shows provider=cohere; ffprobe 6.24s / 790×560 / non-zero frames; ≥6 frames extracted, f_04 content-verified per §11.4.107; output distinct from deepseek/gemini/zipu) | Confirmed:/Volumes/T7/Downloads/Recordings/helix_translate-provider-cohere-translate-wave3b-20260616.mp4 || Mistral | Mistral client | Library | Implemented | Wired | OpenAI-compatible wrapper | Not-inventoried | Source-confirmed | PENDING || xAI | xAI (Grok) client | Library | Implemented | Wired | OpenAI-compatible wrapper | Not-inventoried | Source-confirmed | PENDING || Replicate | Replicate client | Library | Implemented | Wired | Async prediction-polling API | Not-inventoried | Source-confirmed | PENDING || Cerebras | Cerebras client | Library | Implemented | Wired | OpenAI-compatible wrapper (https://api.cerebras.ai/v1, model llama3.1-70b) | Not-inventoried | **PASS — real EN→ES translation captured** (via unified-translator -provider cerebras -model llama3.1-70b: “The morning sun rose over the quiet village.” → “El sol de la mañana se alzó sobre el tranquilo pueblo.” — distinct from SambaNova’s output, proving a genuine per-provider model; Wave-3a, asciinema→agg→ffmpeg, ffprobe 4.03s/4 frames + 11 extracted frames, ≥6 frames content-verified, start-frame ≠ output-frame per §11.4.107) | Confirmed:/Volumes/T7/Downloads/Recordings/helix_translate-provider-cerebras-translate-wave3a-20260616.mp4 || Cloudflare | Cloudflare Workers AI client | Library | Implemented | Wired | /client/v4 Workers AI | Not-

inventoried | Source-confirmed | PENDING | | SiliconFlow | SiliconFlow client | Library | Implemented | Wired | OpenAI-compatible wrapper (specifics UNCONFIRMED) | Not-inventoried | Source-confirmed | PENDING | | Hyperbolic | Hyperbolic client | Library | Implemented | Wired | OpenAI-compatible wrapper (<https://api.hyperbolic.xyz/v1>, model meta-llama/Meta-Llama-3.1-70B-Instruct) | Not-inventoried | **PASS — real EN→ES translation captured** (via unified-translator -provider openai -base-url <https://api.hyperbolic.xyz/v1> -model meta-llama/Meta-Llama-3.1-70B-Instruct: “The morning sun rose over the quiet village. An old fisherman walked along the shore...” → “Al amanecer, el sol salía sobre el tranquilo pueblo. Un anciano pescador caminaba por la orilla...” — distinct verbs (salía/caminaba) from Fireworks’s output (ascendió/paseaba), proving a genuine per-provider model; Wave-3c, asciinema→agg→ffmpeg, ffprobe 8.01s/790×560 yuv420p + 16 extracted frames at fps=2, ≥6 frames content-verified, start-frame (input only) ≠ output-frame (Spanish) per §11.4.107) | Confirmed:/Volumes/T7/Downloads/Recordings/helix_translate-provider-hyperbolic-translate-wave3c-20260616.mp4 | | TogetherAI | TogetherAI client | Library | Implemented | Wired | OpenAI-compatible wrapper (specifics UNCONFIRMED) | Not-inventoried | Source-confirmed | PENDING | | SambaNova | SambaNova client | Library | Implemented | Wired | OpenAI-compatible wrapper (<https://api.sambanova.ai/v1>, model Meta-Llama-3.1-70B-Instruct) | Not-inventoried | **PASS — real EN→ES translation captured** (via unified-translator -provider sambanova -model Meta-Llama-3.1-70B-Instruct: “The morning sun rose over the quiet village.” → “El sol matutino se alzó sobre el tranquilo pueblo.” — distinct from Cerebras’s output, proving a genuine per-provider model; Wave-3a, asciinema→agg→ffmpeg, ffprobe 4.39s/4 frames + 12 extracted frames, ≥6 frames content-verified, start-frame ≠ output-frame per §11.4.107) | Confirmed:/Volumes/T7/Downloads/Recordings/helix_translate-provider-sambanova-translate-wave3a-20260616.mp4 | | Kimi | Kimi (Moonshot) client | Library | Implemented | Wired | OpenAI-compatible wrapper (specifics UNCONFIRMED) | Not-inventoried | Source-confirmed | PENDING | | Novita | Novita client | Library | Implemented | Wired | /v3/ openai-compatible wrapper | Not-inventoried | Source-confirmed | PENDING | | NLPCloud | NLPCloud client | Library | Implemented | Wired | OpenAI-compatible wrapper (specifics UNCONFIRMED) | Not-inventoried | Source-confirmed | PENDING | | Upstage | Upstage (Solar) client | Library | Implemented | Wired | OpenAI-compatible wrapper (specifics UNCONFIRMED) | Not-inventoried | Source-confirmed | PENDING | | Sarvam | Sarvam client | Library | Implemented | Wired | OpenAI-compatible wrapper (specifics UNCONFIRMED) | Not-inventoried | Source-confirmed | PENDING | | Modal | Modal client | Library | Implemented | Wired | OpenAI-compatible wrapper (specifics UNCONFIRMED) | Not-inventoried | Source-confirmed | PENDING | | PublicAI | PublicAI client | Library | Implemented | Wired | OpenAI-compatible wrapper (specifics UNCONFIRMED) | Not-inventoried | Source-confirmed | PENDING | | NIA | NIA client | Library | Implemented | Wired | OpenAI-compatible wrapper (specifics UNCONFIRMED) | Not-inventoried | Source-confirmed | PENDING | | Vulavula | Vulavula client | Library | Implemented | Wired | OpenAI-compatible wrapper (specifics UNCONFIRMED) | Not-inventoried | Source-confirmed | PENDING |

pkg/translator — translation core

Component	Feature	Category	Implementation	Wiring	Real-use
Translator Core	Translator interface + BaseTranslator	Library	Implemented	Wired	Translate/ TranslateWithProgress GetStats + base impl v caching & stats
Translator Core	Translation config	Library	Implemented	Wired	TranslationConfig (provider/model/key/term tokens/lang-pair/script)
Translator Core	UniversalTranslator	Library	Implemented	Wired	High-level ebook translator: lang detection, per-chapter translation, progress events

pkg/ebook — format parsers & writers

Component	Feature	Category	Implementation	Wiring	Real-use
ebook	Universal parser dispatcher	Library	Implemented	Wired	Multi-format parser (FB2/EPUB/...
ebook	FB2 parse	Library	Implemented	Wired	Parse FictionBook 2.0 → Book, n
ebook	FB2 write	Library	Implemented	Wired	Write Book → FB2 XML
ebook	EPUB parse	Library	Implemented	Wired	Parse EPUB (ZIP+OPF/NCX)
ebook	EPUB write	Library	Implemented	Wired	Write Book → EPUB 2.0 ZIP
ebook	DOCX parse	Library	Implemented	Wired	Parse .docx (archive/zip + encodi
ebook	HTML parse	Library	Implemented	Wired	Parse HTML → Book
ebook	TXT parse	Library	Implemented	Wired	Parse plain text (configurable 64
ebook	PDF text extraction	Library	Implemented	Wired	Extract text via ledongthuc/pdf
ebook	DOCX write (output)	Library	Implemented	Wired	Pure-Go OOXML writer pkg/eb archive/zip + encoding/xml, zero OOXML parts; XML specials aut chardata. Wired into unified-trans docx). Real DeepSeek EN→ES - Microsoft Word 2007+, well-f the real translation. Guards: pkg/e subtest in cmd/unified-translator/ proven)

Component	Feature	Category	Implementation	Wiring	Real-use
ebook	PDF write (output)	Library	Implemented	Wired	PDF writer pkg/ebook/pdf_wri renders the in-memory Book to U weasyprint to emit a VALID, C fonts can't render Cyrillic — wea silent-glyph-drop bluff). Wired in generateOutput (case .pdf → e main.go:1160); .pdf is a supporte typed ErrWeasyPrintUnavailabl (§11.4.3/§11.4.6). Guards: pkg/eb (TestPDFWriter_ProducesVali _UnavailableWeasyPrint)

pkg/fb2 — FB2-specific XML

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
fb2	FB2 parser (mixed-content)	Library	Implemented	Wired	Parse FB2 preserving paragraphs/ verses/ inline mixed content	Not-inventoried
fb2	FB2→Markdown converter	Library	Implemented	Wired	Convert FB2 → Markdown (author/ annotation/ epigraph/ poem/cite)	Not-inventoried

pkg/format

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Val
format	Format detector	Library	Implemented	Wired	Detect format by magic bytes/ extension/ content; ZIP disambiguation (EPUB/ DOCX/ AZW3)	Not-inventoried	Sou con

pkg/markdown — EPUB↔Markdown workflow

Component	Feature	Category	Implementation	Wiring	Real-use
markdown	EPUB→Markdown converter	Library	Implemented	Wired	EPUB→MD w/ YAML image extraction, GFM
markdown	Markdown→EPUB converter	Library	Implemented	Wired	MD→EPUB 2.0 w/ OP tables/blockquotes
markdown	Simple workflow	Library	Implemented	Wired	ebook→markdown→tra pipeline w/ progress
markdown	Markdown translator	Library	Implemented	Wired	Line-by-line MD transla preserving formatting/c

pkg/script

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
script	Cyrillic↔Latin conversion	Library	Implemented	Wired	Bidirectional Serbian script conversion, context-aware digraphs	Not-inventoried

pkg/storage

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Val
storage	Storage interface	Library	Implemented	Wired	Abstract session + translation-cache persistence	Not-inventoried	So cor
storage	PostgreSQL backend	Library	Implemented	Wired	Pooled Postgres w/ UNIQUE-tuple dedup	Not-inventoried	So cor
storage	Redis backend	Library	Implemented	Wired	Redis cache, sha256-hashed tuple keys (collision-proof)	Not-inventoried	So cor
storage		Library	Implemented	Wired			

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Val
storage	SQLite backend	Library	Implemented	Wired	SQLite (optional SQLCipher) w/ UNIQUE-tuple migrations	Not-inventoried	So
	Cache-key hashing				Length-prefixed tuple encoding for collision-resistant keys	Not-inventoried	So

pkg/security

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Valida
security	JWT auth service	Library	Implemented	Wired	JWT generate/validate/refresh w/ timing-attack mitigation	Not-inventoried	Source confirm
security	API-key store	Library	Implemented	Wired	Thread-safe key validation w/ expiry & revocation	Not-inventoried	Source confirm
security	Rate limiter	Library	Implemented	Wired	Token-bucket per-key limiting w/ cleanup & stats	Not-inventoried	Source confirm
security	User auth service	Library	Implemented	Wired	Login/user-create w/ timing & enumeration mitigation	Not-inventoried	Source confirm

pkg/verification — quality verification & polishing

Component	Feature	Category	Implementation	Wiring	Real-use
verification	Heuristic verifier	Library	Implemented	Wired	Untranslated-block/HTML-artifact detection, quality score
verification	Multi-pass polisher	Library	Implemented	Wired into unified-translator (-multipass, d53e085)	Iterative multi-LLM polishing, note-taking & consensus. Not reachable via the unified-translator opt-in -multipass flag (+ -multipass-passes -multipass-db; default 0, zero overhead/behavior change when absent). NOTE: distinct from -verify (the CLI's own per-step output check) — -multipass invokes this polisher verification engine. Guaranteed cmd/unified-translator/multipass_wiring_test.go (httptest + real DeepSeek). §11.4.150 research: docs/research/20260615_video_surfaced_verification_multipass_wiring
verification	Book polisher	Library	Implemented	Wired	Multi-LLM per-dimension scoring & consensus validation
verification	Polishing reporter	Library	Implemented	Wired	Stats aggregation (consensus rates, issue/provider breakdown)
verification	Polishing database	Library	Implemented	Wired	SQLite persistence for multiple sessions/pass records
verification	Literary notes	Library	Implemented	Wired	Character/tone/theme/terminology/cultural note collection

internal/verifier — LLMsVerifier consumer integration

Consumer-side LLMsVerifier verification gate (client + scoring + discovery + selection). Source: internal/verifier/pipeline.go.

Component	Feature	Category	Implementation	Wiring	Real-use
internal/verifier	Affirmative-response hard-gate	Library	Implemented	Wired (verification pipeline)	Correctness fix (commit 97a8afd): a model with affirmative_response= is now a HARD disqualification — rejected by the

Component	Feature	Category	Implementation	Wiring	Real-use
					verification gate regardless of other scores

pkg/preparation — pre-translation analysis

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	V
preparation	Preparation coordinator	Library	Implemented	Wired	Multi-pass content analysis (character/terminology/cultural)	Not-inventoried	S c
preparation	Preparation-aware translator	Library	Implemented	Wired	Wraps base translator w/ analysis phase + context passing	Not-inventoried	S c
preparation	Prompt builder	Library	Implemented	Wired	Context-aware analysis prompts refined across passes	Not-inventoried	S c
preparation	Domain types	Library	Implemented	Wired	Content-classification/character/cultural/terminology structs	Not-inventoried	S c
preparation	Utilities	Library	Implemented	Wired	JSON serialization + human-readable summaries	Not-inventoried	S c

pkg/coordination — multi-LLM coordination

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
coordination	Multi-LLM coordinator	Library	Implemented	Wired	Multiple LLM instances w/ round-robin, retry, distributed fallback	Not inventoried
coordination	Multi-LLM translator wrapper	Library	Implemented	Wired	Adapts coordinator to translator.Translator interface	Not inventoried

pkg/distributed — SSH distributed workers

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
distributed	Coordinator (distribute + aggregate)	Library	Implemented	Wired	Distributes work across remote LLM instances, round-robin, failover to local	Not-inventoried
distributed	SSH connection pool	Library	Implemented	Wired	Pooled SSH connections w/ idle-timeout cleanup	Not-inventoried
distributed	Worker pairing/discovery	Library	Implemented	Wired	Discover/pair/health-check remote translator services	Not-inventoried
distributed	Fallback / graceful degradation	Library	Implemented	Wired	Retry w/ backoff, degraded-mode entry/exit, recovery tracking	Not-inventoried
distributed	Version manager	Library	Implemented	Wired	Version check/update/rollback,	Not-inventoried

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
distributed	Performance instrumentation	Library	Implemented	Wired	signed packages, backups, drift alerts (no LLM) Conn pooling, TTL result cache, circuit breaker, batch processing	Not-inventoried
distributed	Security hardening	Library	Implemented	Wired	SSH host-key verify, TLS validation, network ACLs, security event log	Not-inventoried
distributed	Manager / orchestration	Library	Implemented	Wired	Init/discover/pair/version-manage + translation delegation	Not-inventoried

pkg/sshworker

Obsolete-Details: Since=bridge phase-2 R-2/R-4 (2026-06-16); Reason=feature-removed (local-runtime + SSH-local translation removed; default path now sources the LLMsVerifier bridge — no local llama.cpp/Ollama, no SSH-local worker); Superseding-item=pkg/bridge (LLMsVerifier→agent bridge); Operator-approval=2026-06-15 D1/D2 (forbid all local runtimes) + R-1 redirect + the R-4 “keep distributed API, remove SSH-local” decision (2026-06-16); Triple-check evidence=source files removed (FACT: `git ls-files` empty for the removed `cmd/*` dirs + `pkg/sshworker` + `pkg/modelsbridge` + `ollama.go/llamacpp*.go`) + CM-NO-LOCAL-RUNTIME gate PASS. The KEPT distributed-API path is NOT Obsolete. | Component | Feature | Category | Implementation | Wiring | Real-use | Tests | Validation | Video-confirmation | |-----|-----|-----|-----|-----|-----|-----|-----|-----|-----| sshworker | SSH worker | Library | **Obsolete (→ Fixed.md)** | Wired | SSH exec, file up/download, codebase sync, key generation | Not-inventoried | Source-confirmed | N/A | | sshworker | Monitored worker | Library | **Obsolete (→ Fixed.md)** | Wired | Event-monitored progress tracking for long commands | Not-inventoried | Source-confirmed | N/A |

pkg/deployment

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
deployment	Deployment orchestrator	Library	Implemented	Wired	Automated multi-host instance deployment	Not-inventoried
deployment	Docker orchestrator	Library	Implemented	Wired	Generate/manage docker-compose for multi-service deploy	Not-inventoried
deployment	SSH deployer	Library	Implemented	Wired	SSH-based deploy w/ validation & error handling	Not-inventoried
deployment	Network discovery	Library	Implemented	Wired	UDP service discovery/broadcast	Not-inventoried
deployment	API comm logger	Library	Implemented	Wired	Retrofit-style REST request/response logging between nodes	Not-inventoried
deployment	Deployment types	Library	Implemented	Wired	DeploymentPlan/Status/NetworkService/APICommunicationLog	Not-inventoried

pkg/grpc

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
grpc	gRPC TranslationService	Library	Implemented	Wired	gRPC server: sessions, event streaming, provider registry, concurrency	Not-inventoried
grpc	Core translator pipeline	Library	Implemented	Wired	Translation job exec w/ step tracking + progress via llm factory	Not-inventoried

pkg/api (library layer)

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Value
api	REST API server	Library	Implemented	Wired	HTTP server: auth, CSRF,	Not-inventoried	Scalable, secure, easy to use

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Val
api	Translation endpoints	Library	Implemented	Wired	rate-limit, routing text/FB2/ ebook translate, batch ops, lang/provider listing	Not-inventoried	Sc co
api	Batch translation handlers	Library	Implemented	Wired	Batch requests w/ queue, progress, concurrency	Not-inventoried	Sc co
api	Verifier endpoints	Library	Implemented	Wired	LLMsVerifier model verification/ scoring/ selection; / verified-models	Not-inventoried	Sc co

pkg/events, websocket, batch, progress, report, logger, hardware, hash

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Val
events	Event pub/sub bus	Library	Implemented	Wired	Thread-safe pub/sub w/ subscription IDs + type filtering	Not-inventoried	Sou con
websocket	WebSocket hub	Library	Implemented	Wired	Real-time session event broadcast to dashboards	Not-inventoried	Sou con
batch	Batch processor	Library	Implemented	Wired	File/dir translation w/ concurrency & progress	Not-inventoried	Sou con
progress	Progress tracker	Library	Implemented	Wired	Chapter/section progress w/ ETA	Not-inventoried	Sou con
report		Library	Implemented	Wired			

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Val
logger	Report generator	Library	Implemented	Wired	Session reporting (issues/warnings/logs/results)	Not-inventoried	Sou
	Structured logger				Text/JSON logging w/ levels	Not-inventoried	Sou
	Hardware detector				CPU/RAM/GPU detection + model-size estimation	Not-inventoried	Sou
	Codebase hash generator				SHA256 of source/config/deps/binaries	Not-inventoried	Sou

pkg/language

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Validation
language	Language registry / heuristic detector	Library	Implemented	Wired	ISO 639-1 registry + heuristic detection (18+ langs)	Not-inventoried	Source-confirmed
language	LLM language detector	Library	Implemented	Wired	LLM-driven language detection	Not-inventoried	Source-confirmed

pkg/models & modelsbridge

Obsolete-Details: Since=bridge phase-2 R-2/R-4 (2026-06-16); Reason=feature-removed (local-runtime + SSH-local translation removed; default path now sources the LLMsVerifier bridge — no local llama.cpp/Ollama, no SSH-local worker); Superseding-item=pkg/bridge (LLMsVerifier→agent bridge); Operator-approval=2026-06-15 D1/D2 (forbid all local runtimes) + R-1 redirect + the R-4 “keep distributed API, remove SSH-local” decision (2026-06-16); Triple-check evidence=source files removed (FACT: `git ls-files` empty for the removed cmd/* dirs +

pkg/sshworker + pkg/modelsbridge + ollama.go/llamacpp*.go) + CM-NO-LOCAL-RUNTIME gate PASS. The KEPT distributed-API path is NOT Obsolete. | Component | Feature | Category | Implementation | Wiring | Real-use | Tests | Validation | Video-confirmation | |---| | models | Model registry | Library | Implemented | Wired | Translation-model metadata registry (50+ models) | Not-inventoried | Source-confirmed | N/A | | models | Model downloader | Library | Implemented | Wired | HTTP model download w/ caching + sha256 verification | Not-inventoried | Source-confirmed | N/A | | models | User domain | Library | Implemented | Wired | User struct, bcrypt hashing, in-memory repo | Not-inventoried | Source-confirmed | N/A | | models | Error definitions | Library | Implemented | Wired | Domain error constants | Not-inventoried | Source-confirmed | N/A | | models | Verifier registry | Library | Implemented | Wired | LLMsVerifier-backed registry w/ thresholds & filtering | Not-inventoried | Source-confirmed | PENDING | | modelsbridge | Models bridge | Library | **Obsolete (→ Fixed.md)** | Wired | Adapter mapping models.LLMRequest/Response → llm.LLMClient | Not-inventoried | Source-confirmed | N/A |

pkg/bridge — LLMsVerifier→component+agent bridge

In-process bridge (no local models) that selects the best LLMsVerifier-scored model and routes component/agent calls to it; backs cmd/model-bridge CLI + MCP. Source: pkg/bridge/bridge.go, commit ab1bed3 (+ routing mutation seam a5860b2).

Component	Feature	Category	Implementation	Wiring
bridge	Best-model selection	Library	Implemented	Wired (via cmd/model-bridge CLI+MCP)
bridge	Invoke routes to BestModel	Library	Implemented	Wired
bridge	BestTranslator (single strongest translator + fallback chain)	Library	Implemented	Wired (R-1 redirect seam)

Component	Feature	Category	Implementation	Wiring
bridge	BestTranslatorFunc (context-only single-translator closure)	Library	Implemented	Wired (R-1 redirect seam)
bridge	EnsembleFactory (provider-diverse ensemble seam)	Library	Implemented	Wired (R-1 ensemble redirect)
bridge	ProviderDiverseTranslators	Library	Implemented	Wired (backs EnsembleFactory)
bridge	ProviderDiverseClients	Library	Implemented	Wired
bridge	ProviderDiverseModels	Library	Implemented	Wired (backs the ProviderDiverse* set)

Component	Feature	Category	Implementation	Wiring
bridge	BestClient (single strongest low-level client)	Library	Implemented	Wired

pkg/version

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Validation
version	App version	Library	Implemented	Wired	Singleton version constant (2.3.0)	Not-inventoried	Source confirmed
version	Codebase hasher	Library	Implemented	Wired	SHA256 codebase versioning over dirs/extensions w/ exclude patterns	Not-inventoried	Source confirmed

pkg/challenge_runner — anti-bluff test harness (uses digital.vasic.challenges)

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
challenge_runner	Challenge orchestrator	Library	Implemented	Wired	Coordinates 18+ integration challenges, parallel exec + filtering	Not-inventoried
challenge_runner	Build & format-detection challenges	Library	Implemented	Wired	Build-verification + FB2/EPUB/TXT/HTML/PDF/DOCX format-detection challenges	Not-inventoried
challenge_runner	Infrastructure challenges	Library	Implemented	Wired	Event-bus + other infra challenges	Not-inventoried

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
challenge_runner	Provider challenges	Library	Implemented	Wired	(run go test) Verify LLM provider factory/ interface compile+load (build, not live calls)	Not-inventori

3. Web Dashboards & Endpoints

Web Dashboards

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
monitor.html	WebSocket connection status	Web	Implemented	Wired	Green/red pulse dot + Connected/Disconnected indicator	Not-inver
monitor.html	Session/Client ID display	Web	Implemented	Wired	Static fields (demo-session-123, web-demo)	Not-inver
monitor.html	Overall progress bar + percent	Web	Implemented	Wired	Driven by WS translation_progress/step_completed events	Not-inver
monitor.html	Current step / step message panel	Web	Implemented	Wired	Shows current step + message from WS	Not-inver
monitor.html	Duration timer	Web	Implemented	Wired	Client-side elapsed clock	Not-inver
monitor.html	Status badge	Web	Implemented	Wired	Pending/Running/Completed/Failed	Not-inver
monitor.html	Event log	Web	Implemented	Wired	Scrolling list of received WS events	Not-inver
monitor.html	Control Panel: Simulate Progress	Web	Implemented	Wired	Locally fakes & echoes progress events over WS (demo only)	Not-inver

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
monitor.html	Clear Log / Disconnect / Reconnect WS	Web	Implemented	Wired	Local log clear + WS connect/disconnect to ws://localhost:8090/ws	Not-inver
enhanced-monitor.html	Stats header (Active/Completed/Errors)	Web	Implemented	Wired	Aggregated counters from WS-tracked sessions	Not-inver
enhanced-monitor.html	Control panel: Connect/Disconnect/Clear	Web	Implemented	Wired	WS to ws://localhost:8090/ws?client_id=dashboard	Not-inver
enhanced-monitor.html	Monitor specific session	Web	Implemented	Wired	Input + button to filter UI to a given session_id	Not-inver
enhanced-monitor.html	Worker Information panel	Web	Implemented	Wired	Host:port, model, capacity from WS worker_info	Not-inver
enhanced-monitor.html	Current Session Progress panel	Web	Implemented	Wired	Progress bar, step, current/total items, duration, status	Not-inver
enhanced-monitor.html	Live progress chart (Chart.js)	Web	Implemented	Wired	Progress % over time, last 20 points	Not-inver
enhanced-monitor.html	Event log	Web	Implemented	Wired	Color-coded WS event stream	Not-inver
enhanced-monitor.html	Session History table	Web	Implemented	Wired	Last 10 sessions: id, worker, status, duration, progress, start	Not-inver
dashboard.html	Stats cards (Total/Completed/Running/Failed)	Web	Implemented	Wired	Polls GET /api/v1/translations every 5s	Not-inver
dashboard.html	Active Sessions table	Web	Implemented	Wired	Lists sessions w/ provider, status, progress, duration	Not-inver
dashboard.html	Session Details view	Web	Implemented	Wired	Progress bar, steps, generated files via GET /api/v1/translations/:id	Not-inver
dashboard.html		Web	Implemented	Wired	Connects ws://localhost:8080/ws?session_id=...;	Not-inver

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
	Live progress via WebSocket				progress_update/step_*	
dashboard.html	New Translation modal	Web	Implemented	Wired	File input, provider selector, source/target lang, API-key/SSH/llama fields (UI only)	Not-inver
dashboard.html	Start Translation (submit)	Web	Implemented	Wired	POST /api/v1/translations with provider config — triggers backend translation	Not-inver
dashboard.html	Cancel session	Web	Implemented	Wired	DELETE /api/v1/translations/:id	Not-inver

Web Dashboard backend handler — pkg/api/dashboard.go

Backend that serves the dashboard page (/, /dashboard, /monitor) and the translations endpoints. Was previously UNROUTED (returned 404); now wired so the dashboard genuinely translates for end users. Source: pkg/api/dashboard.go, commit f6ba5cc.

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
api/ dashboard.go	Dashboard page + translations endpoints	Web	Implemented	Wired (was Unwired/404 pre-fix)	Serves the dashboard HTML at /, /dashboard, /monitor and the GET/POST /api/v1/translations path; Start-Translation now runs a real translation through the wired UI path (no longer a 404)	Covered + RED → GR guard dashboard (default GR asserts page real translat RED_MODE= reproduces fix 404) — go test -pkg/api G this session

REST API — pkg/api/server.go (standalone Server, gin)

Component	Feature	Category	Implementation	Wiring	Real-use
api/server.go	GET /health	API	Implemented	Wired	Status + translator name
api/server.go	POST /api/translate	API	Implemented	Wired	translator.Translate(...)
api/server.go	GET /api/languages	API	Implemented	Wired	Static language list
api/server.go	GET /api/stats	API	Implemented	Wired	Returns hardcoded zeros (not live metrics)
api/server.go	POST /api/upload	API	Implemented	Wired	Save uploaded file, detect format
api/server.go	POST /api/batch	API	Partial	Wired	Returns batch_id queued only; performs NO translation

REST API — pkg/api/handler.go (primary Handler, gin)

Component	Feature	Category	Implementation	Wiring	Real-use
api/handler.go	GET /health, /, /api/v1/version, /api/v1/providers, /api/v1/stats	API	Implemented	Wired	Static/info cache stats
api/handler.go	GET /ws	API	Implemented	Wired	WebSocket to wsHub
api/handler.go	POST /api/v1/translate	API	Implemented	Wired	Distributed Translation target_l was hardcoded

Component	Feature	Category	Implementation	Wiring	Real-use
api/handler.go	POST /api/v1/translate/fb2	API	Implemented	Wired	Parse ebook
api/handler.go	POST /api/v1/translate/batch	API	Implemented	Wired	Loops text
api/handler.go	POST /api/v1/convert/script	API	Implemented	Wired	Cyrillic ^[2]
api/handler.go	GET /api/v1/status/:session_id	API	Implemented	Wired	Returns ha a live sess
api/handler.go	GET /api/v1/languages	API	Implemented	Wired	Static list language.C
api/handler.go	POST /api/v1/translate/validate	API	Implemented	Wired	Validates n length); no
api/handler.go	POST /api/v1/preparation/ analyze	API	Stub	Wired	File stat/co analysis, m
api/handler.go	GET /api/v1/preparation/ result/:session_id	API	Stub	Wired	Returns m
api/handler.go	POST /api/v1/translate/ebook	API	Stub	Wired	Emits star does NOT
api/handler.go	POST /api/v1/translate/ cancel/:session_id	API	Implemented	Wired	Emits can cancelled
api/handler.go	GET /api/v1/distributed/status	API	Implemented	Wired	dm.GetSt
api/handler.go	POST /api/v1/distributed/ workers/discover	API	Implemented	Wired	SSH work
api/handler.go	POST /api/v1/distributed/ workers/:worker_id/pair	API	Implemented	Wired	Pair work
api/handler.go	DELETE /api/v1/distributed/ workers/:worker_id/pair	API	Implemented	Wired	Unpair wo
api/handler.go	POST /api/v1/distributed/ translate	API	Implemented	Wired	dm.Trans
api/handler.go	POST /api/v1/update/ {upload,apply,rollback}	API	Implemented	Wired	Worker se install/rol
api/handler.go	GET /api/v1/monitoring/version/ {metrics,alerts,health,dashboard}	API	Implemented	Wired	Version-sy
api/handler.go	POST /api/v1/monitoring/ version/drift-check	API	Implemented	Wired	Trigger dr
api/handler.go	GET /api/v1/monitoring/version/ alerts/history	API	Implemented	Wired	Alert histo
api/handler.go	POST /api/v1/monitoring/ version/alerts/:alert_id/ acknowledge	API	Implemented	Wired	Ack alert
api/handler.go	POST /api/v1/monitoring/ version/alerts/channels/ {email,webhook,slack}	API	Implemented	Wired	Add alert
		Web	Implemented	Wired	Serve emb

Component	Feature	Category	Implementation	Wiring	Real-use
api/ handler.go	GET /api/v1/monitoring/version/ dashboard.html				
api/ handler.go	POST /api/v1/auth/login, /auth/ token	API	Implemented	Wired	JWT auth
api/ handler.go	GET /api/v1/profile	API	Implemented	Wired	Return cla middleware

REST API — pkg/api/batch_handlers.go (under /api/v1)

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
api/ batch_handlers.go	POST / api/v1/ translate/ string	API	Implemented	Wired	NewLLMTranslator + Translate	Not- invent
api/ batch_handlers.go	POST / api/v1/ translate/ directory	API	Implemented	Wired	Batch processor translates each file	Not- invent

REST API — pkg/api/verifier_handlers.go (LLMsVerifier, under /api/v1)

Component	Feature	Category	Implementation	Wiring	Real-use
api/ verifier_handlers.go	GET /api/ v1/verified- models	API	Implemented	Wired (only when LLMsVerifier enabled)	List verified models (LLMsVerifier API call). GAP (§11.4.6): LLMsVerifier is DISABLED in the current config, so the route is not mounted and returns HTTP 404. Not a defect — feature- disabled-by- config; needs a verifier- enabled

Component	Feature	Category	Implementation	Wiring	Real-use
					config to confirm
api/verifier_handlers.go	GET /api/v1/verified-models/:id	API	Implemented	Wired	Get one verified model
api/verifier_handlers.go	GET /api/v1/verification-status	API	Implemented	Wired	Ping LLMsVerifier
api/verifier_handlers.go	POST /api/v1/verification/refresh	API	Implemented	Wired	Refresh scores
api/verifier_handlers.go	GET /api/v1/providers/verified	API	Implemented	Wired	Aggregated verified providers
api/verifier_handlers.go	POST /api/v1/translate-with-verification	API	Stub	Wired	Selects/verifies a model, returns metadata + text preview; does NOT translate

REST API — cmd/api-server/main.go (gRPC-proxying REST, :8080)

Component	Feature	Category	Implementation	Wiring	Real-use
api-server	POST /api/v1/translations	API	Implemented	Wired	Proxies grpcClient.StartTranslating
api-server	GET /api/v1/translations/:session_id	API	Implemented	Wired	Get status (proxy)
api-server	GET /api/v1/translations	API	Implemented	Wired	List (proxy)
api-server	DELETE /api/v1/translations/:session_id	API	Implemented	Wired	Cancel (proxy)
api-server	GET /api/v1/translations/:session_id/stream	API	Implemented	Wired	WS proxy of gRPC StreamTranslationP
api-server	GET /api/v1/providers	API	Implemented	Wired	Get providers (prox
api-server	GET /api/v1/health	API	Implemented	Wired	Derived from gRPC state
api-server	GET /api/v1/metrics	API	Implemented	Wired	

Component	Feature	Category	Implementation	Wiring	Real-use
					Returns static/zero r (not live)
api-server	GET /ws	Web	Implemented	Wired	WS upgrade, register wsHub (requires session_id)
api-server	GET /static/*	Web	Implemented	Wired	Static file server (./v static)
api-server	GET /	Web	Implemented	Wired	Render dashboard.h template

WebSocket — hub & fan-out

Component	Feature	Category	Implementation	Wiring	Real-use	Tests	Valid
websocket/ hub.go	/ws endpoint (package mux)	Infra	Implemented	Wired	WS upgrade; reads client_id + session_id	Not- inventoried	Sourc confi
websocket/ hub.go	Event fan-out	Infra	Implemented	Wired	Subscribes to ALL bus events; marshals events.Event JSON; per- session filter; Broadcast()	Not- inventoried	Sourc confi

gRPC — pkg/grpc/server.go (service TranslationService)

Component	Feature	Category	Implementation	Wiring	Real-use
grpc	StartTranslation	gRPC	Implemented	Wired	Reserves session spawns runTrans → translator.Transla
grpc	GetTranslationStatus	gRPC	Implemented	Wired	Snapshot session translator.GetSta
grpc	ListTranslations	gRPC	Implemented	Wired	Iterate sessions
grpc	CancelTranslation	gRPC	Implemented	Wired	

Component	Feature	Category	Implementation	Wiring	Real-use
grpc	StreamTranslationProgress	gRPC	Implemented	Wired	Cancel ctx + translator.Cancel Stream session progress events
grpc	GetProviders	gRPC	Implemented	Wired	Static ProviderRegistry (openai, anthropic, ssh)
grpc	SubscribeEvents	gRPC	Implemented	Wired	Stream filtered SystemEvents from bus

CORS / Middleware / Auth (Infra)

Component	Feature	Category	Implementation	Wiring	Real-use
cmd/server	corsMiddleware	Infra	Implemented	Wired	Driven by cfg.Security.CORSOrigins; reflects allowlisted Origins + Allow-Credentials; wildcard * no-credentials OPTIONS→204
cmd/server	rateLimitMiddleware	Infra	Implemented	Wired	Per-ClientIP via security.RateLimiter; 404 on exceed
pkg/api/handler.go	authMiddleware	Infra	Implemented	Wired	Bearer JWT validation applied to protected subgroup, gated by Security.EnableAuth
pkg/api/server.go	authMiddleware (standalone)	Infra	Implemented	Wired	X-API-Key header == config.Security.APIKey; Security.RequireAuth
pkg/grpc/server.go	message-size limits	Infra	Implemented	Wired	4 MB recv/send msg-sizes; no interceptors/auth/rate-limit registered

4. Owned Submodules

challenges/

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
challenges	Challenge core engine	Submodule	Implemented	Wired	Challenge interface, Config, Result, BaseChallenge, ProgressReporter (liveness)	Not-inver
challenges	Registry + dependency ordering	Submodule	Implemented	Wired	Challenge registration, Kahn topological sort by deps	Not-inver
challenges	Runner / execution engine	Submodule	Implemented	Wired	Sequential/parallel/ pipeline execution, liveness/stall monitoring	Not-inver
challenges	Assertion engine	Submodule	Implemented	Wired	16 built-in evaluators (not_empty, contains, min_count, max_latency...) + custom	Not-inver
challenges	Report generation	Submodule	Implemented	Wired	Markdown / JSON / HTML reporters	Not-inver
challenges	Live monitoring	Submodule	Implemented	Wired	WebSocket dashboard, progress liveness	Not-inver
challenges	Metrics	Submodule	Implemented	Wired	Prometheus-compatible challenge metrics	Not-inver
challenges	HTTP client	Submodule	Implemented	Wired	Generic REST client with JWT auth, functional options	Not-inver
challenges	Challenge bank loader	Submodule	Implemented	Wired	Load challenge definitions from JSON/YAML	Not-inver
challenges	Plugin system	Submodule	Implemented	Wired	Custom challenge types + assertion plugins	Not-inver
challenges	Infra bridge	Submodule	Implemented	Wired	Bridge to digital.vasic.containers for test infra	Not-inver
challenges	User-flow automation	Submodule	Implemented	Wired	8 adapter interfaces / 21 impls (Playwright, ADB, Tauri, HTTP, gRPC, WS, Gradle, Cargo, npm), 19	Not-inver

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
					templates, 12 evaluators	
challenges	userflow-runner CLI	Submodule	Implemented	Wired	CLI runner for user-flow challenges	Not-inventoried
challenges	panoptic / i18n / env / logging	Submodule	Implemented	Wired	Observation, i18n, env-var redaction, structured logging	Not-inventoried
challenges	anti_bluff lib	Submodule	Implemented	Wired	Shell anti-bluff helper library	Not-inventoried

containers/

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
containers	boot manager	Submodule	Implemented	Wired	On-demand infra boot/start orchestration	Not-inventoried
containers	compose	Submodule	Implemented	Wired	docker/podman compose file orchestration	Not-inventoried
containers	health checking	Submodule	Implemented	Wired	TCP/HTTP health-check primitives	Not-inventoried
containers	runtime auto-detect	Submodule	Implemented	Wired	Docker/ Podman/ Kubernetes runtime detection	Not-inventoried
containers	endpoint	Submodule	Implemented	Wired	ServiceEndpoint builder	Not-inventoried
containers	lifecycle	Submodule	Implemented	Wired	Container lifecycle management	Not-inventoried
containers	service discovery / registry	Submodule	Implemented	Wired	Service discovery + registry	Not-inventoried
containers	orchestrator / scheduler	Submodule	Implemented	Wired	Orchestration + scheduling	Not-inventoried
containers	emulator suite	Submodule	Implemented	Wired	Android/device emulator boot/ matrix/cleanup/ canary	Not-inventoried
containers	genymotion	Submodule	Implemented	Wired	Genymotion device support	Not-inventoried
containers	vm-matrix	Submodule	Implemented	Wired		

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
					VM matrix orchestration	Not-inventoried
containers	distributed build/test	Submodule	Implemented	Wired	Distributed build + test commands	Not-inventoried
containers	crossbuild	Submodule	Implemented	Wired	Cross-platform build	Not-inventoried
containers	ctop	Submodule	Implemented	Wired	Container top / monitoring TUI	Not-inventoried
containers	network / volume	Submodule	Implemented	Wired	Network + volume primitives	Not-inventoried
containers	supporting primitives	Submodule	Implemented	Wired	cache/metrics/monitor/event/policy/remote/lazyservice/brokertest/envconfig	Not-inventoried

doc_processor/

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
doc_processor	Document loader	Submodule	Implemented	Wired	Loads Markdown / YAML / HTML docs (10 MB cap)	Not-inventoried
doc_processor	Feature-map builder	Submodule	Implemented	Wired	Builds structured FeatureMap; heuristic extraction + LLM Enrich	Not-inventoried
doc_processor	LLMAgent interface	Submodule	Implemented	Wired	Summarize/ExtractFeatures/ClassifyFeature/InferScreens/GenerateTestSteps	Not-inventoried
doc_processor	Coverage tracker	Submodule	Implemented	Wired	Verification coverage tracking by platform/category	Not-inventoried
doc_processor	DocGraph	Submodule	Implemented	Wired	Doc node/edge graph	Not-inventoried
doc_processor	Config / i18n	Submodule	Implemented	Wired	Configuration + i18n	Not-inventoried
doc_processor		Submodule	Implemented	Wired		

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
	docprocessor CLI				CLI running feature extraction against a docs dir	Not- inven
doc_processor	archdoc (internal)	Submodule	Implemented	Wired	Architecture-doc helper (LLM-use UNCONFIRMED)	Not- inven

docs_chain/

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
docs_chain	sync command	Submodule	Implemented	Wired	Propagate changes through chain (regenerate + re-export atomically)	Not- inventori
docs_chain	verify command	Submodule	Implemented	Wired	Deterministic CI/ pre-build gate (chain in sync)	Not- inventori
docs_chain	doctor command	Submodule	Implemented	Wired	Diagnostics	Not- inventori
docs_chain	graph command	Submodule	Implemented	Wired	Show DAG / topo order	Not- inventori
docs_chain	watch daemon	Submodule	Implemented	Wired	fsnotify-driven continuous sync	Not- inventori
docs_chain	exec: transforms	Submodule	Implemented	Wired	Shell-out transforms (pandoc md→html/docx, weasyprint html→pdf)	Not- inventori
docs_chain	content- hash engine	Submodule	Implemented	Wired	Content-hash incremental recompute; roster/ corpus fingerprint (§11.4.86)	Not- inventori
docs_chain	DAG + recompute	Submodule	Implemented	Wired	Node/Edge/ Graph, cycle detection, Kahn topo, early-cutoff, sync-edge conflict	Not- inventori
docs_chain	adapters	Submodule	Implemented	Wired	FileAdapter, SQLiteAdapter (pure-Go modernc),	Not- inventori

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
docs_chain	orchestrator	Submodule	Implemented	Wired	DERIVED pandoc/ weasyprint (ToolAbsentError) Atomic run: staged writes, SQLite-txn commit + rollback	Not-inventori
docs_chain	DB builtins	Submodule	Implemented	Wired	Bidirectional md- to-sqlite / sqlite- to-md (row-level drift); colorize- html	Not-inventori
docs_chain	config + state	Submodule	Implemented	Wired	YAML multi- context loader; state.json	Not-inventori

llm_orchestrator/

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
llm_orchestrator	Agent interface + AgentPool	Submodule	Implemented	Wired	Thread-safe pool, capability matching, acquire/ release, health	Not-invent
llm_orchestrator	CLI agent adapters	Submodule	Implemented	Wired	OpenCode, ClaudeCode, Gemini, Junie, QwenCode adapters (BaseAdapter)	Not-invent
llm_orchestrator	Transport / protocol	Submodule	Implemented	Wired	PipeTransport (JSON-line stdin/stdout) + FileTransport, protocol types	Not-invent
llm_orchestrator	Response parser	Submodule	Implemented	Wired	Parses agent JSON-line responses	Not-invent
llm_orchestrator	HealthMonitor + CircuitBreaker	Submodule	Implemented	Wired	Health tracking + fault tolerance over agents	Not-invent

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
llm_orchestrator	Config / i18n	Submodule	Implemented	Wired	Pool/agent config; i18n	Not-invent
llm_orchestrator	orchestrator CLI	Submodule	Implemented	Wired	CLI entry for orchestration	Not-invent

llm_provider/

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
llm_provider	LLMProvider interface	Submodule	Implemented	Wired	Complete/CompleteStream/HealthCheck/GetCapabilities/ValidateConfig	Not-inven
llm_provider	Circuit breaker	Submodule	Implemented	Wired	closed/open/half-open fault tolerance	Not-inven
llm_provider	Health monitor	Submodule	Implemented	Wired	healthy/degraded/unhealthy/unknown	Not-inven
llm_provider	Retry logic	Submodule	Implemented	Wired	Exponential backoff + jitter, HTTP-status detection	Not-inven
llm_provider	Model discovery	Submodule	Implemented	Wired	Provider/model discovery (HTTP/endpoint-based)	Not-inven
llm_provider	API-key management	Submodule	Implemented	Wired	API key handling	Not-inven
llm_provider	HTTP client + models + i18n	Submodule	Implemented	Wired	Shared HTTP client, request/response models, i18n	Not-inven
llm_provider	Provider implementations (43)	Submodule	Implemented	Wired	ai21, anthropic, cerebras, chutes, claude, cloudflare, codestral, cohere, deepseek, fireworks, gemini, generic, githubmodels, groq, huggingface, hyperbolic,	Not-inven

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
-----------	---------	----------	----------------	--------	----------	-------

junie, kilo, kimi,
mistral, modal,
nia, nlpcloud,
novita, nvidia,
ollama, openai,
openrouter,
perplexity,
publicai, qwen,
replicate,
sambanova,
sarvam,
siliconflow,
together,
upstage, venice,
vulavula, xai,
zai, zen, zhipu
(source
llm_provider/
pkg/
providers/
<name>/)

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
					LLM-powered summarization	
llms_verifier	LLMOps / self-improve / benchmark	Submodule	Implemented	Wired	LLM operations, self-improvement loop, benchmarking	Not-invented
llms_verifier	Messaging	Submodule	Implemented	Wired	Notification/messaging integration (Splunk/DataDog/etc.)	Not-invented
llms_verifier	suffix / verified export	Submodule	Implemented	Wired	Mandatory branding suffix + verified-only config export	Not-invented
llms_verifier	Auth / security / API / API-keys	Submodule	Implemented	Wired	LDAP/SSO/SAML/OIDC, SQLCipher, REST API, key mgmt	Not-invented
llms_verifier	Multi-platform clients	Submodule	Implemented	Wired	CLI, TUI, Web, Desktop, Mobile (UI; LLM-backed)	Not-invented
llms_verifier	helixqa / cliagents / crush / opencode	Submodule	Implemented	Wired	Integration packages (LLM-use UNCONFIRMED)	Not-invented
llms_verifier	scheduler / partners / bigdata / multimodal / performance	Submodule	Implemented	Wired	Scheduling, partners, big-data, multimodal, perf (multimodal likely LLM)	Not-invented
llms_verifier	main CLI + ACP	Submodule	Implemented	Wired	Main CLI; ACP (Agent Client Protocol)	Not-invented

security/

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
security	Content guardrails	Submodule	Implemented	Wired	Rule pipeline (MaxLength, ForbiddenPatterns, RequireFormat) w/ severities	Not-invented
security	PII detection + redaction	Submodule	Implemented	Wired	Email/Phone/SSN/CreditCard(Luhn)/	Not-invented

Component	Feature	Category	Implementation	Wiring	Real-use	Tests
					IPv4 detection; Mask/Hash/Remove	
security	Content filtering	Submodule	Implemented	Wired	Composable chain: Length/Pattern/Keyword filters	Not-invented
security	Policy enforcement	Submodule	Implemented	Wired	Named policies, 8 condition operators, Allow/Deny/Audit + audit trail	Not-invented
security	Vulnerability scanner	Submodule	Implemented	Wired	Findings/reports, severity filtering + merging	Not-invented
security	HTTP security headers	Submodule	Implemented	Wired	Middleware: CSP/HSTS/X-Frame-Options + 7 standard headers	Not-invented
security	Secure storage	Submodule	Implemented	Wired	AES-256-GCM encrypted file KV store, Argon2id KDF	Not-invented
security	Privilege-escalation scanner	Submodule	Implemented	Wired	Probes /proc/self/status, capability bitmasks, namespace cgroups	Not-invented
security	SSRF denial	Submodule	Implemented	Wired	Denies SSRF on caller's outbound HTTP client	Not-invented
security	E2EE	Submodule	Implemented	Wired	End-to-end encryption	Not-invented
security	Attestation / GPU attestation	Submodule	Implemented	Wired	Attestation primitives	Not-invented
security	i18n	Submodule	Implemented	Wired	i18n	Not-invented

vision_engine/

Component	Feature	Category	Implementation	Wiring	Real-use
vision_engine	Analyzer interface	Submodule	Implemented	Wired	UI-element/text-region issue detection, screen diff, key frames

Component	Feature	Category	Implementation	Wiring	Real-use
vision_engine	LLM vision adapters	Submodule	Implemented	Wired	OpenAI GPT-4o, Anthr Claude, Gemini 2.0 Fla Qwen VL, Kimi, StepF Ollama + Fallback chain
vision_engine	NavigationGraph	Submodule	Implemented	Wired	Directed graph of screen transitions
vision_engine	Remote Ollama deployment	Submodule	Implemented	Wired	SSH auto-install, mode lifecycle; GPU/CPU/R detection; distributed IL RPC
vision_engine	OpenCV (stub + build-tag real)	Submodule	Partial	Wired	OpenCV-based CV — s default, real impl behind tag
vision_engine	Config / i18n	Submodule	Implemented	Wired	Configuration (HELIX_VISION_PRO etc.), i18n

Honest gaps & caveats (§11.4.6)

1. **Video coverage is 30 feature rows.** 30 features are confirmed by real, ffprobe- and content-verified recordings (Wave 2 of 2026-06-16 added 5 net-new — DOCX output, PDF output, HTML→Markdown conversion, ProviderDiverseTranslators, EnsembleFactory — plus an additional re-confirming clip on the already-confirmed gRPC TranslationService row, no double-count; Wave 1 added 2 — bridge-ensemble best-model selection + bridge provider-diverse catalogue; see the Anti-bluff note + the Coverage-summary Video—confirmed row for the authoritative count and full list — reconciled with both, §11.4.60). DOCX output and PDF output, formerly real-artifact-only, are now video-confirmed by Wave-2 recordings. Every other runtime/user-visible feature is PENDING a recording or N/A. This is stated, not hidden. 1a. **DOCX output is Implemented (Rev 3, commit 87cd2be)** — pure-Go OOXML writer; a real run produced a Microsoft Word 2007+ .docx; **now video-confirmed (Wave 2)** by helix_translate-docx-output-20260616-112943.mp4 (openai/gpt-4, real Spanish in the .docx). **PDF output is Implemented (Rev 4, commit fb265e7)** — pkg/ebook/pdf_writer.go (Book→HTML5→weasyprint→Cyrillic-faithful PDF), wired into unified-translator (.pdf output, main.go:1160); pkg/ebook PDF tests green; **now video-confirmed (Wave 2)** by helix_translate-pdf-output-20260616-113057.mp4 (openai/gpt-4, valid PDF with real Spanish). Output formats are EPUB/FB2/TXT/HTML/MD/DOCX/PDF. 1b. **-multipass ≠ -verify.** The unified-translator -verify flag runs the CLI's own per-step output check; the separate -multipass flag (d53e085) invokes the pkg/verification multi-pass polisher engine — now wired and video-confirmed. 1c. **cmd/translator, cmd/translate-ssh, cmd/ebook-translator are now Obsolete (→ Fixed.md), Rev 7** — RECONCILED with the per-binary table

- rows (lines ~136/218/233) which mark all three **Obsolete** (→ **Fixed.md**) with Reason=feature-removed. The local-runtime + SSH-local translation paths were REMOVED in bridge phase-2 R-2..R-4 (operator-approved 2026-06-15 D1/D2 + R-4 “keep distributed API, remove SSH-local”); the default path now sources the LLMsVerifier bridge (pkg/bridge). These are therefore NO LONGER open STUB / OPERATOR-BLOCKED items — the earlier “STUB / OPERATOR-BLOCKED” wording was stale and is corrected here. The KEPT distributed-API translation path is NOT Obsolete. 1e. **GET /api/v1/verified-models returns 404** — LLMsVerifier is disabled in the current config so the route is not mounted (feature-disabled-by-config, not a defect). 1f. **cmd/cli title path-leak FIXED** (d53e085) + **write-safety FIXED** (87cd2be, atomic write/no-clobber, guard cmd/cli/no_partial_output_test.go).
2. **Tests = Not-inventoried** for all rows: the raw inventory did not perform per-feature test attribution. The repo carries extensive `_test.go` + `test/` suites, but mapping each to a feature is future work — recorded as honest unknown, never guessed Covered/None.
 3. **4 STUB endpoints** return mock data and do NOT do the named work: `POST /api/v1/translate/ebook`, `POST /api/v1/preparation/analyze`, `GET /api/v1/preparation/result/:id`, `POST /api/v1/translate-with-verification`.
 4. **1 Partial endpoint:** standalone `pkg/api/server.go` `POST /api/batch` returns a queued `batch_id` only and performs no translation.
 5. **Static/hardcoded returns** (not stubs, but not live): `GET /api/stats` (zeros), `GET /api/v1/status/:id` (hardcoded completed), `GET /api/v1/metrics` (static), monitor-server `GET /api/v1/status` (static monitoring_active), several static language/provider lists. Marked Implemented with the static behaviour stated in Real-use.
 6. **vision_engine OpenCV** is Partial — stub by default, real CV only behind a build tag.
 7. **UNCONFIRMED LLM determinations** carried verbatim from the inventory for thin OpenAI-compatible providers, several `llms_verifier` packages, challenges/userflow evaluators, and `doc_processor/archdoc` — noted inline rather than asserted.
 8. **✅ Wave-1 DISCOVERED BLOCKER — RESOLVED (Wave 2, §11.4.4 / §11.4.107 / §11.4.108 / §11.4.138).** The default bridge TRANSLATOR-path runtime failure described below is FIXED: the Wave-2 recordings (DOCX output, PDF output, gRPC translate, and the 4 per-provider unified clips novita/mistral/groq/siliconflow) are real-use videos validator-verified (ffprobe+frame) showing the verified-bridge translator path producing real target-language output — the exact path that failed in Wave 1 now works, which is why those rows flip to CONFIRMED this wave. The historical forensic record of the blocker is preserved below for §11.4.6 / §11.4.138 traceability. **[HISTORICAL — Wave-1 finding, now fixed]** While recording Wave 1 (DOCX/PDF/gRPC), a real run of unified-translator (and, by source, the gRPC backend `pkg/grpc/core_translator.go` which uses `BestTranslatorFunc`) FAILS at translate time with:
bridge: no verified translator available: model 'Sao10K/L3-8B-Stheno-v3.2' is not valid for provider 'novita'. Valid models: [meta-llama/llama-3.1-70b-instruct mistralai/mixtral-8x7b-instruct]. **Root cause (FACT, §11.4.6):**
VerifiedFactory.CreateTranslatorWithFallback / CreateTranslator build the verified translator via NewLLMTranslatorWithConfig, which re-validates the model against the STALE hardcoded ValidModels whitelist (`pkg/translator/llm/llm.go:227-241`); a genuinely LLMsVerifier-verified

model whose ID is not in that whitelist is rejected. The `BestModel/``invoke` path (used by `model-bridge` `invoke` + Wave-1 videos) correctly BYPASSES this whitelist (its own code comment, `bridge.go:250-258`, documents exactly why a verified model must not be gated on the whitelist) — and so does NOT exhibit the bug. **Impact:** unified-translator (all outputs: TXT/HTML/EPUB/FB2/MD/DOCX/PDF) + gRPC `translate` are non-functional via the default bridge path until fixed. This BLOCKS Wave-1 videos #2 (DOCX), #3 (PDF), #4 (gRPC), and the per-provider unified-path; recording any of them as “working” would be a §11.4.107 bluff, so they are honest SKIPS this wave. **Fix direction (not yet applied — needs its own §11.4.115 RED→GREEN + §11.4.142 independent review + §11.4.145 impact research + §11.4.40 full-suite retest):** route `VerifiedFactory`’s verified-translator construction through the `ResolvedProvider` resolver (`FactoryProvider`=“openai” + `BaseURL`) the way `bridge.realClientBuild` already does, bypassing the `ValidModels` whitelist for already-verified models. Tracked as a Wave-2 / dedicated fix item (severity HIGH — release-critical default path).